PRIME POLYNOMIALS IN LEGENDRE INTERVALS OVER $\mathbb{F}_{2^K}[T]$, III: THE HEXAD OF A QUARTIC LINEARISED POLYNOMIAL, THE SYMPLECTIC GROUP $\mathrm{Sp}_4(\mathbb{F}_2)$, AND THE WITT SUMS BEHIND THE CONSTANT X

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ABSTRACT. Let $q = 2^k$ with k odd. Part II of this series reduced the second of the three constants governing the variance of prime polynomials in Legendre intervals at degree $d = 6$, $X = \sum_{\alpha \in U} \chi(e_2(\alpha))$ with $U = \{\alpha \in \mathbb{F}_{q^{12}} : \mathrm{Tr}\,\alpha = \mathrm{Tr}\,\alpha^3 = \mathrm{Tr}\,\alpha^5 = 0\}$, to sums $W'(\lambda_3, \lambda_5)$ of characters of order 4 of the Hayes group, whose fine structure is governed by the kernel of the linearised polynomial $L'_\lambda(x) = \lambda_5^4 x^{16} + \lambda_3^4 x^8 + x^4 + \lambda_3^2 x^2 + \lambda_5 x$. Numerically, the Frobenius acted on this four-dimensional $\mathbb{F}_2$-space through $\mathrm{Sp}_4(\mathbb{F}_2) \cong S_6$ with exact polynomial class frequencies, and W' was a function of the class. This paper explains the phenomenon and proves a substantial part of it. The kernel of L'_λ is the set of pairwise sums of the six roots of the sextic

$$B_\lambda(u) = \lambda_5^2 u^6 + \lambda_3^2 u^2 + \lambda_5 u + 1 = u\,A_\lambda(u) + 1,$$

where A_λ is the Bluher quintic whose roots span the kernel of the quadratic-character analogue $L_\lambda = L'_\lambda - x^4$; the proof is the identity $L'_\lambda = x^4 B_\lambda^2 + B_\lambda + 1$ together with a division of B_λ by a quadratic factor. Consequently the Frobenius permutes the six roots, the S_6-class is the factorisation type of B_λ over $\mathbb{F}_q$, the symplectic form is the duad–syntheme pairing of the generalised quadrangle $W(2)$, the roots are six of the sixteen elements of the coset $T'^{-1}_\lambda(1)$ of the polar operator, and the one-parameter deformation $L_\lambda + \varepsilon x^4$ corresponds to $uA_\lambda(u) + \sqrt{\varepsilon}$, interpolating between the quadratic sums of A_6 and the order-4 sums of X. We prove the first moment identities of the family (the numbers of roots in $\mathbb{F}_q$, of ordered pairs of roots in $\mathbb{F}_q$, and of roots in $\mathbb{F}_{q^2} \setminus \mathbb{F}_q$ total $(q-1)^2$, $(q-2)^2$ and $q(q-2)$) and verify the eleven class counts up to $q = 512$. The order-4 sums are then treated as Frobenius traces of rank-four Witt–Artin–Schreier sheaves. Using the hexad, a multiplicity formula for Witt traces along towers, and the moduli of Gauss sums of $\mathbb{Z}/4$-valued quadratic forms, we prove: every individual sum $S''(\lambda)$ is a rational integer; the normalised Frobenius eigenvalues are roots of unity; in the class of 5-cycles the L-polynomial is $(1 - \eta^5 q^{5/2} T^5)/(1 - \eta q^{1/2} T)$ with η a *primitive* 8th root of unity, so that $S''(\lambda) = -q^6$ for every λ and $W' = -q^7$, the term carrying the leading constant of X; in the six classes whose kernel lies in $\mathbb{F}_{q^4}$ the sum over $\mathbb{F}_{q^{12}}$ equals q^4 times the sum over $\mathbb{F}_{q^4}$; in the remaining classes every $S''(\lambda)$ equals $\pm 4q^6$. The closed form $X = q^6 + q^5$ is thereby reduced to the class counts and to the Brown invariants of explicit $\mathbb{Z}/4$-valued forms over $\mathbb{F}_{q^4}$ and $\mathbb{F}_{q^6}$, all stated precisely and verified for $q \le 128$.

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1. Introduction

1.1. **Background.** Let $q = 2^k$, $d \geq 2$, $n = 2d$, and let $\mathcal{G} = \mathcal{G}_{d-1} = \{1 + c_1 u + \cdots + c_{d-1}u^{d-1}\}$ be the Hayes group of truncated power series over $\mathbb{F}_q$, which classifies monic polynomials of degree n by their top $d-1$ coefficients. The Legendre interval of a monic f of degree d is $\mathcal{I}_f = \{f^2 + s : \deg s \leq d\}$, the fibre over the class of f^2; in characteristic 2 these classes form the subgroup $\mathcal{G}^2$ of squares, of index $q^{\lfloor d/2 \rfloor}$, and the variance of the prime-power count $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ over Legendre intervals is amplified by a factor $\asymp q^2$ at $d = 4$ and $d = 6$ relative to the Keating–Rudnick value for the full group [4, 5]. At $d = 6$ Part II [5] proved that Ψ takes exactly three values on the q^2 Legendre classes

and that the variance is an explicit function of three constants

$$A_6 = |U|, \qquad X = \sum_{\alpha\in U} \chi(e_2(\alpha)), \qquad Y = \sum_{\alpha\in U} \chi(e_4(\alpha)),$$
$$U = \{\alpha \in \mathbb{F}_{q^{12}} : \operatorname{Tr}\alpha = \operatorname{Tr}\alpha^3 = \operatorname{Tr}\alpha^5 = 0\},$$

where $\chi = (-1)^{\operatorname{Tr}_{q/2}}$ and $e_j(\alpha)$ is the j-th elementary symmetric function of the twelve conjugates of α over $\mathbb{F}_q$; explicitly (Part II, §8)

$$\operatorname{Var}_{\mathcal{G}^2}(\Psi) = \frac{(A_6 - q^9)^2 + (q-1)X^2}{q^4} + \frac{(q-1)Y^2}{q^3}.$$

The constant A_6 was evaluated for all odd k ($A_6 = q^9 + (q-1)q^5$): it is a sum of quadratic-character sums $S(\lambda) = \sum_{x\in\mathbb{F}_{q^{12}}} (-1)^{\operatorname{Tr}(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)}$, i.e. of Frobenius traces of supersingular Artin–Schreier curves, and the sum over λ_1 of the $S(\lambda)$ with fixed (λ_3, λ_5) is governed by the Frobenius action on the kernel of the linearised polynomial

$$L_\lambda(x) = \lambda_5^4 x^{16} + \lambda_3^4 x^8 + \lambda_3^2 x^2 + \lambda_5 x,$$

which is the sum-zero $\mathbb{F}_2$-span of the five roots of the Bluher quintic $A_\lambda(u) = \lambda_5^2 u^5 + \lambda_3^2 u + \lambda_5$, so that the Frobenius acts through S_5 and only three cycle types occur (a parity argument); the 4-cycles give $2q^7$, the rest 0.

The constant X is a sum of characters of order 4: by Part II, §6.7,

$$X = q^{-3} \sum_{\lambda_3,\lambda_5\in\mathbb{F}_q} W'(\lambda_3,\lambda_5), \qquad W'(\lambda_3,\lambda_5) = \sum_{\lambda_1\in\mathbb{F}_q} S'(\lambda_1,\lambda_3,\lambda_5), \tag{1}$$

where $S'(\lambda) = \sum_{x\in\mathbb{F}_{q^{12}}} (-1)^{\operatorname{Tr}(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x) + \operatorname{Tr}_{q/2} e_2(x)}$, and the role of L_λ is taken by

$$L'_\lambda(x) = L_\lambda(x) + x^4 = \lambda_5^4 x^{16} + \lambda_3^4 x^8 + x^4 + \lambda_3^2 x^2 + \lambda_5 x. \tag{2}$$

The extra term x^4, which comes from the polar form of $\operatorname{Tr}_{q/2} e_2$, destroys both the quintic factorisation of the kernel equation and the scaling symmetry $(\lambda_3,\lambda_5) \mapsto (s^3\lambda_3, s^5\lambda_5)$ that made W a function of the single invariant $j = \lambda_3^5/\lambda_5^3$. Part II could therefore only record what the exact computations showed (Table 3 there): for every $(\lambda_3,\lambda_5) \in (\mathbb{F}_q^\times)^2$ the Frobenius acts on the four-dimensional kernel $\mathcal{K}_\lambda = \ker L'_\lambda$ through one of the eleven conjugacy classes of $S_6 \cong \mathrm{Sp}_4(\mathbb{F}_2)$ (never through the classes of order 7 or 15 of $\mathrm{GL}_4(\mathbb{F}_2) \cong A_8$), the number of pairs in each class is an exact polynomial in q, and W' depends only on the class. The closed form $X = q^6 + q^5$ remained a conjecture (Conjecture 6.19 there), and the first of the open problems listed at the end of Part II asked for the sextic whose six roots have the kernel as their pairwise sums. Throughout this paper $\lambda_5 \neq 0$; the terms of (1) with $\lambda_5 = 0$ (the pure cubic sums and $W'(0,0)$) were evaluated in Part II, Lemmas 6.15–6.16, and enter only in Corollary 1.5.

1.2. **Main results.** The sextic exists and is as simple as one could hope.

Theorem 1.1 (the hexad)**.** *Let $q = 2^k$ be arbitrary, $\lambda_3 \in \mathbb{F}_q$, $\lambda_5 \in \mathbb{F}_q^\times$, and*

$$B_\lambda(u) = \lambda_5^2 u^6 + \lambda_3^2 u^2 + \lambda_5 u + 1 = u\,A_\lambda(u) + 1.$$

Then:

(a) $L'_\lambda(x) = x^4 B_\lambda(x)^2 + B_\lambda(x) + 1$ *identically.*
(b) B_λ *is separable, with roots $r_1,\dots,r_6 \in \overline{\mathbb{F}}_2$ satisfying $e_1 = e_2 = e_3 = 0$ and $r_i \neq 0$, and the fifteen sums $r_i + r_j$ $(i<j)$ are distinct and nonzero.*
(c) $\mathcal{K}_\lambda = \ker L'_\lambda = \{0\} \cup \{r_i + r_j : i<j\}$. *For distinct roots r, s with $x = r+s$, $p = rs$ one has $B_\lambda(x) = (\lambda_5 p x)^2 + 1$ and $\lambda_5^2 p^2 x^3 + x + \lambda_5 p = 0$.*

(d) *The polar operator* $T'_\lambda(x) = \lambda_5 x^4 + \lambda_3 x^2 + x + (\lambda_3 x)^{1/2} + (\lambda_5 x)^{1/4}$, *which satisfies* $L'_\lambda = (T'_\lambda)^4$, *takes the value* 1 *at every root:* $T'_\lambda(r_i) = 1$. *The coset* $T'^{-1}_\lambda(1) = r_1 + \mathcal{K}_\lambda$ *consists of the six roots and the ten* synthematic totals $r_i + r_j + r_k = r_l + r_m + r_n$, *and* $x \in T'^{-1}_\lambda(1)$ *is a root of* B_λ *iff* $x^4 B_\lambda(x) \neq 1$.

The identification of a four-dimensional $\mathbb{F}_2$-space with the pairwise sums of a six-element set with vanishing sum (a *hexad*) is the classical construction of the symplectic space $\mathbb{F}_2^4$ with its 15 points (duads) and 15 totally isotropic lines (synthemes), on which $\mathrm{Sp}_4(\mathbb{F}_2) \cong S_6$ acts [6]; the symplectic form is $\omega(r_i + r_j, r_k + r_l) = |\{i,j\} \cap \{k,l\}| \bmod 2$. Since B_λ has coefficients in $\mathbb{F}_q$, the q-Frobenius permutes its roots and hence acts on $\mathcal{K}_\lambda$ through S_6; this is the explanation of the observations of Part II.

Theorem 1.2 (Frobenius classes)**.** *Let* σ *be the* q*-Frobenius. The action of* σ *on* $\mathcal{K}_\lambda$ *is the action of the permutation* $\pi_\lambda \in S_6$ *of the roots of* B_λ *on duads, and the conjugacy class of* π_λ, *i.e. the factorisation type of* B_λ *over* $\mathbb{F}_q$, *determines: the orbit structure of* σ *on* $\mathcal{K}_\lambda \setminus 0$, *the field* $\mathbb{F}_q(\mathcal{K}_\lambda) = \mathbb{F}_{q^{N_0}}$, *the fixed spaces of* σ^m *on* $\mathcal{K}_\lambda$ *for all* m, *and the radicals of all the quadratic forms of Section 6 (Table 1). All eleven classes of* S_6 *occur for* $q \geq 32$, *and for* $(\lambda_3, \lambda_5) \in (\mathbb{F}_q^\times)^2$ *the orbit structures are exactly the eleven patterns of Part II, Table 3.*

The canonical hexad is one of several that the Frobenius preserves: σ lies in $28|t|/|C|$ of the 28 conjugates of $\mathrm{Sp}_4(\mathbb{F}_2)$ in $\mathrm{GL}_4(\mathbb{F}_2)$ with type t, where C is its $\mathrm{GL}_4(\mathbb{F}_2)$-class (Proposition 4.3); in particular a Frobenius of order 4 is of type $4\,1^2$ for one of its two invariant symplectic forms and of type $4\,2$ for the other, and only the canonical form, the one defined by B_λ, predicts W'.

The one-parameter family $L_\lambda + \varepsilon x^4$ interpolates between the two situations of Part II: its kernel is the pairwise-sum set of the roots of $uA_\lambda(u) + \sqrt{\varepsilon}$ (Theorem 3.7); at $\varepsilon = 0$ the hexad degenerates to $\{0\} \cup \{\text{roots of } A_\lambda\}$ and S_6 to the stabiliser S_5 of the root 0, which is the quintic frame of Part II.

For the counting problem we obtain the first moments.

Theorem 1.3 (moments of the sextic family)**.** *Over all* $(\lambda_3, \lambda_5) \in \mathbb{F}_q \times \mathbb{F}_q^\times$ *(equivalently over all monic sextics* $u^6 + au^2 + bu + b^2$, $b \neq 0$*):*

$$\sum N_1(B_\lambda) = (q-1)^2, \qquad \sum N_1(B_\lambda)\big(N_1(B_\lambda) - 1\big) = (q-2)^2,$$
$$\sum \big(N_2(B_\lambda) - N_1(B_\lambda)\big) = q(q-2),$$

where $N_m(B)$ *is the number of roots of* B *in* $\mathbb{F}_{q^m}$. *Moreover any two distinct roots* r, s *of one* B_λ *that are both in* $\mathbb{F}_q$, *or conjugate over* $\mathbb{F}_q$ *in* $\mathbb{F}_{q^2}$, *satisfy* $\mathrm{Tr}_{q/2}(r+s) = 0$.

The eleven class counts of Part II (reproduced in Table 3) are compatible with these identities and are now a statement about an explicit family of sextics; we verified them for all $q \leq 512$ (Proposition 5.2), and the counts of the sub-family $\lambda_3 = 0$ are theorems. Their proof in general is left open (Conjecture 5.3).

The second half of the paper concerns the sums themselves. The natural objects are not the $S'(\lambda)$ of (1) but the genuine order-4 sums

(3)
$$S''(\lambda) = \sum_{x \in \mathbb{F}_{q^{12}}} i^{\mathfrak{q}_\lambda(x)}, \qquad \mathfrak{q}_\lambda(x) = \mathrm{Tr}_{W_2(\mathbb{F}_{q^{12}})/\mathbb{Z}/4}[x] + 2\,\mathrm{Tr}_{q^{12}/2}(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x) \in \mathbb{Z}/4,$$

which agree with the $S'(\lambda)$ after summation over λ_1 (so $W' = \sum_{\lambda_1} S''$) and are the traces of Frobenius on rank-four Witt–Artin–Schreier sheaves over $\mathbb{F}_q$. The function $\mathfrak{q}_\lambda$ is a $\mathbb{Z}/4$-valued quadratic form on $\mathbb{F}_{q^{12}}$ whose polar form is $2\,\mathrm{Tr}(x\,T'_\lambda(y))$ and whose radical is

$\mathcal{K}_\lambda \cap \mathbb{F}_{q^{12}}$; its analogues $\mathfrak{q}_\lambda^{(m)}$ on $\mathbb{F}_{q^m}$ give the sums $s_m(\lambda)$ over $\mathbb{F}_{q^m}$, and $s_m = -\sum_{j=1}^4 \omega_j^m$ for the four Frobenius eigenvalues ω_j of the sheaf. Write $N_0 = [\mathbb{F}_q(\mathcal{K}_\lambda) : \mathbb{F}_q]$ (Table 1).

Theorem 1.4 (Witt sums through the hexad)**.** *Let k be odd and $(\lambda_3, \lambda_5) \in \mathbb{F}_q \times \mathbb{F}_q^\times$.*

(a) (Integrality.) *$S''(\lambda) \in \mathbb{Z}$ for every λ_1, and $S''(\lambda_1, \lambda_3, \lambda_5) = S''(\lambda_1 + 1, \lambda_3, \lambda_5)$. If $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{12}}$ then $S''(\lambda) \in \{0, \pm 4q^6\}$; otherwise (the class of 5-cycles) $S''(\lambda) = \pm q^6$.*

(b) (Roots of unity.) *$|\omega_j| = \sqrt{q}$ and the $\omega_j^{4N_0}$ are all equal to $-s_{4N_0}(\lambda)/4$, which is q^{2N_0} times an 8th root of unity. Hence the $\omega_j/\sqrt{q}$ are roots of unity, and $s_m(\lambda) = -\sum_j \omega_j^m$ for all m is determined by $s_1, \dots, s_4$.*

(c) (The class 5 1.) *If B_λ has an irreducible quintic factor over $\mathbb{F}_q$, then $s_m(\lambda) = s_1(\lambda)^m$ for $5 \nmid m$, $s_1(\lambda) = \eta\sqrt{q}$ with η a primitive 8th root of unity, the L-polynomial of the sheaf is $(1 - \eta^5 q^{5/2} T^5)/(1 - \eta q^{1/2} T)$, and*

$$S''(\lambda_1, \lambda_3, \lambda_5) = -q^6 \quad \text{for every } \lambda_1, \qquad W'(\lambda_3, \lambda_5) = -q^7.$$

(d) (The classes with $N_0 \mid 4$: 1^6, $2\,1^4$, $2^2 1^2$, 2^3, $4\,1^2$, $4\,2$.) *$S''(\lambda) = q^4\, s_4(\lambda)$, and $s_4(\lambda) \in \{0, \pm 4q^2\}$. If $N_0 \mid 2$ then $s_4(\lambda) \neq 0$ for every λ_1; if $N_0 = 4$ then $s_4(\lambda) \neq 0$ for exactly $q/2$ values of λ_1 or for none.*

(e) (The classes with $N_0 \in \{3, 6\}$: $3\,1^3$, 3^2, $3\,2\,1$, 6.) *$S''(\lambda) = \pm 4q^6$ for every λ_1. If $N_0 = 3$ and $s_3(\lambda) \neq 0$ then $S''(\lambda) = +4q^6$. If $s_6(\lambda) \neq 0$ then $S''(\lambda) = -4q^6\nu_6^2$ where $\nu_6 = -s_6(\lambda)/4q^3 \in \{\pm 1, \pm i\}$. In the classes 3^2 and 6 all q sums $S''(\lambda_1, \lambda_3, \lambda_5)$ are equal, so $W'(\lambda_3, \lambda_5) = \pm 4q^7$.*

Parts (c)–(e) are the mechanism behind the values of W' observed in Part II: $-q^7$ in the class 5 1 (now a theorem), $\pm 2q^7$ in the classes of order 4 ($q/2$ nonzero sums of the same sign), $4q^7$ or 0 elsewhere (q nonzero sums, all of the same sign or half of each). What remains is the *sign* of the Gauss sums s_4 on $\mathbb{F}_{q^4}$ (classes with $N_0 \mid 4$) and s_6 on $\mathbb{F}_{q^6}$ or s_3 on $\mathbb{F}_{q^3}$ (the others), i.e. the Brown invariants of explicit $\mathbb{Z}/4$-valued quadratic forms. We computed them for all λ at $q = 8$, for four pairs per class (all λ_1) at $q = 32$, and, for even k, for all λ at $q = 4$ and $q = 16$: the predicted S'' sum to the exact W' in every case and the normalised eigenvalues are 24th roots of unity (40th in the class 5 1) depending only on the class and on a coset of λ_1 (Tables 5, 6, Figure 4). The resulting description is Conjecture 6.17; with the class counts it gives $X = q^6 + q^5$.

Corollary 1.5. *For odd k the identity $X = q^6 + q^5$, hence the unconditional form of the variance theorem of Part II for odd k, follows from Conjecture 5.3 (the class counts) and Conjecture 6.17 (the signs of the small-field Gauss sums); both are verified exactly for $q \leq 128$, the former for $q \leq 512$. The contribution of the class 5 1, namely $-\frac{q^2+1}{5}q^7$, is unconditional given the count.*

1.3. **Organisation.** Section 2 recalls linearised polynomials, the Bluher quintic, duads and synthemes, Witt vectors and $\mathbb{Z}/4$-valued quadratic forms. Section 3 proves Theorem 1.1 and the deformation theorem. Section 4 proves Theorem 1.2 and tabulates the classes. Section 5 proves Theorem 1.3 and discusses the class counts. Section 6 develops the Witt-sum calculus (the two parametrisations, the $\mathbb{Z}/4$-forms and their radicals, the multiplicity formula along towers, integrality, the eigenvalue theorem) and proves Theorem 1.4 class by class. Section 7 reports the computations. Section 8 lists what remains.

2. Preliminaries

2.1. **Notation.**

$q=2^k$, σ	the ground field size and the q-Frobenius $x\mapsto x^q$
Tr, τ	$\mathrm{Tr}_{q^{12}/q}$ (also written τ); other traces carry their fields as indices
$\lambda=(\lambda_1,\lambda_3,\lambda_5)$, f_λ	$\lambda_5x^5+\lambda_3x^3+\lambda_1x$
L_λ, $L'_\lambda=L_\lambda+x^4$	the linearised polynomials (2); T_λ, $T'_\lambda=T_\lambda+1$ their polar operators (4)
A_λ, $B_\lambda=uA_\lambda+1$	the Bluher quintic and the sextic of the hexad (Theorem 1.1)
$\mathcal{K}^0_\lambda=\ker L_\lambda$, $\mathcal{K}_\lambda=\ker L'_\lambda$	the quintic frame and the hexad space, $\dim_{\mathbb{F}_2}=4$
$r_1,\dots,r_6$; $\pi_\lambda\in S_6$	the roots of B_λ and the permutation induced by σ
N_0, f_m	$[\mathbb{F}_q(\mathcal{K}_\lambda):\mathbb{F}_q]$ and $\dim\mathcal{K}_\lambda^{\sigma^m}$ (Table 1)
$\tau_0=r_5+r_6$	the sum of the two roots outside a 4-cycle (classes $4\,1^2$, $4\,2$)
T_N	the Witt trace $\mathrm{Tr}_{W_2(\mathbb{F}_{2^N})/\mathbb{Z}/4}[x]$, (6)
$\mathfrak{q}^{(m)}_\lambda$, $s_m(\lambda)$	the $\mathbb{Z}/4$-form on $\mathbb{F}_{q^m}$ and its Gauss sum, (11); $\mathfrak{q}_\lambda=\mathfrak{q}^{(12)}_\lambda$, $S''=s_{12}$
$S'(\lambda)$, $W'(\lambda_3,\lambda_5)$	the $\mathbb{F}_2$-form sums of Part II and $\sum_{\lambda_1}S'=\sum_{\lambda_1}S''$, (1)
ω_j, $\zeta_j=\omega_j/\sqrt{q}$	the Frobenius eigenvalues of the rank-four Witt sheaf and their normalisations
V_λ, $\bar{Q}$	$\{c:T'_\lambda(c)\in\mathbb{F}_q\}$ and the phase form (Proposition 6.15)

2.2. **Linearised polynomials and adjoints.** Throughout $q=2^k$, $\bar{\mathbb{F}}=\bar{\mathbb{F}}_2$, and $\mathrm{Tr}=\mathrm{Tr}_{q^{12}/q}$ unless indicated. An $\mathbb{F}_2$-linearised polynomial is $L(x)=\sum_i a_ix^{2^i}$; its kernel in $\bar{\mathbb{F}}$ is an $\mathbb{F}_2$-space of dimension $\deg_2 L$ when L is separable, i.e. when $a_0\neq0$. The adjoint of the $\mathbb{F}_2$-linear map $D(x)=\sum_i a_ix^{2^i}$ on $\mathbb{F}_{q^N}$ with respect to the trace form $\mathrm{Tr}_{q^N/2}(xy)$ is $D^*(y)=\sum_i(a_iy)^{2^{-i}}$; $D+D^*$ is self-adjoint, and the $\mathbb{F}_2$-quadratic form $x\mapsto\mathrm{Tr}_{q^N/2}(xD(x))$ has polar form $\mathrm{Tr}_{q^N/2}(y\,(D+D^*)(x))$. We use

$$D_\lambda(x)=\lambda_5x^4+\lambda_3x^2,\qquad T_\lambda=D_\lambda+D^*_\lambda,\qquad T'_\lambda=T_\lambda+\mathrm{id},\tag{4}$$

so that $T_\lambda(x)^4=L_\lambda(x)$ and $T'_\lambda(x)^4=L'_\lambda(x)$, and $\ker T_\lambda=\ker L_\lambda$, $\ker T'_\lambda=\ker L'_\lambda$ in $\bar{\mathbb{F}}$.

2.3. **The Bluher quintic and the quintic frame.** For $\lambda_5\neq0$ put $A_\lambda(u)=\lambda_5^2u^5+\lambda_3^2u+\lambda_5$. Part II, Proposition 6.7, shows $x^4L_\lambda(x)=R(x)^2+R(x)$ with $R=x^5A_\lambda(x)$, so that $\ker L_\lambda=\{0\}\cup\{A_\lambda=0\}\cup\{x^5A_\lambda(x)=1\}$, that A_λ is separable with $e_1=0$, and that $\mathcal{K}^0_\lambda:=\ker L_\lambda$ is the sum-zero $\mathbb{F}_2$-span of the five roots $u_0,\dots,u_4$ of A_λ: its fifteen nonzero elements are the u_i and the ten sums u_i+u_j. With $u=\lambda_5^{-1/5}t^{1/2}v$ the quintic becomes the Gold polynomial v^5+v+c, $c=(\lambda_5^3/\lambda_3^5)^{1/2}$, whose factorisation types over $\mathbb{F}_q$ for odd k are $(1,4)$, $(2,3)$, $(1,1,1,2)$ with $(q-2)/2$, $(q+1)/3$, $(q-2)/6$ values of c (Part II, Proposition 6.8; [1, 9]). We shall see that $\{0\}\cup\{u_i\}$ is the degenerate hexad.

2.4. **Hexads, duads, synthemes.** Let V be a four-dimensional $\mathbb{F}_2$-space. A *frame* is a five-element set of nonzero vectors summing to 0 any four of which are independent; there are $|\mathrm{GL}_4(\mathbb{F}_2)|/5!=168$ frames. A *hexad* is a six-element subset H of an affine space over V with $\sum_{h\in H}h=0$ (the sum is taken in V after choosing an origin in H, and is independent of the choice since 6 is even) whose fifteen pairwise sums (*duads*) are distinct and nonzero; the hexads containing the origin are $\{0\}\cup F$, F a frame, and two hexads with the same duad set differ by a translation by an element of V. The duads of a hexad are all of $V\setminus0$, and the three pairwise-disjoint duads of a *syntheme* $\{\{i,j\},\{k,l\},\{m,n\}\}$ sum to 0: a syntheme is a two-dimensional subspace. The 15 duads and 15 synthemes with the incidence "duad $\in$ syntheme" form the generalised quadrangle $W(2)$, the duads being the points of $\mathbb{P}(V)$ and the synthemes the totally isotropic lines of the alternating form

$$\omega(r_i+r_j,\ r_k+r_l)=|\{i,j\}\cap\{k,l\}|\bmod2,\tag{5}$$

which is nondegenerate; S_6 acting on the six elements of the hexad acts on V through $\mathrm{Sp}(V,\omega)\cong\mathrm{Sp}_4(\mathbb{F}_2)$, and this is the exceptional isomorphism $S_6\cong\mathrm{Sp}_4(\mathbb{F}_2)$ [6]. Conversely each of the $|\mathrm{GL}_4(\mathbb{F}_2)|/|\mathrm{Sp}_4(\mathbb{F}_2)| = 20160/720 = 28$ nondegenerate alternating forms on V arises from exactly one translation class of hexads (168 frames, six per class). The elements of the coset r_1+V other than the six r_i are the ten sums $r_i+r_j+r_k$ (each equal to the sum of the complementary triple); these are the *synthematic totals*. Figure 1 shows the configuration.

2.5. **Witt vectors of length two and the characters of order four.** For an $\mathbb{F}_2$-algebra R, $W_2(R)=R\times R$ with $(a_0,a_1)+(b_0,b_1)=(a_0+b_0,\,a_1+b_1+a_0b_0)$ and $(a_0,a_1)(b_0,b_1)=(a_0b_0,\,a_0^2b_1+a_1b_0^2)$; $W_2(\mathbb{F}_2)=\mathbb{Z}/4$ via $(a_0,a_1)\mapsto a_0+2a_1$, and the Teichmüller lift is $[x]=(x,0)$. For a finite field $\mathbb{F}_{2^N}$ the trace $\mathrm{Tr}_{W_2(\mathbb{F}_{2^N})/\mathbb{Z}/4}$ is the sum of the N Frobenius conjugates, and for $x\in\mathbb{F}_{2^N}$

$$\mathsf{T}_N(x) := \mathrm{Tr}_{W_2(\mathbb{F}_{2^N})/\mathbb{Z}/4}[x] = \mathrm{Tr}_{2^N/2}(x) + 2\,e_2^{\mathrm{abs}}(x) \in \mathbb{Z}/4, \tag{6}$$

where $e_2^{\mathrm{abs}}(x)=\sum_{0\le i<j<N}x^{2^i+2^j}\in\mathbb{F}_2$, since the Witt sum of Teichmüller lifts $(x_i,0)$ has second component $e_2(x_1,x_2,\dots)$. The identity $[x]+[y]=(x+y,xy)$ gives

$$\mathsf{T}_N(x+y)=\mathsf{T}_N(x)+\mathsf{T}_N(y)+2\,\mathrm{Tr}_{2^N/2}(xy)\qquad(\text{in }\mathbb{Z}/4), \tag{7}$$

so T_N is a $\mathbb{Z}/4$-valued quadratic form whose polar form is twice the trace form. We shall also need the integer multiples of a Teichmüller lift: $2[x]=(0,x^2)$, $3[x]=(x,x^2)$, $4[x]=0$, i.e. $n[x]=n\cdot[x]$ with $n\in\mathbb{Z}/4=W_2(\mathbb{F}_2)$ acting by the ring structure. In the Hayes group $\mathcal{G}_5$ at $d=6$ the subgroup $\mathcal{G}^2=\{(c_2,c_4)\}$ is $W_2(\mathbb{F}_q)$ (Part II, Proposition 8.3), and the q^3 characters of $\mathcal{G}_5$ that restrict to $(-1)^{\mathrm{Tr}\,c_2}$ on $\mathcal{G}^2$ are, on the class of $\alpha\in\mathbb{F}_{q^{12}}$, the functions $i^{\mathsf{T}_{12k}(\lambda_1\alpha)}(-1)^{\mathrm{Tr}(\lambda_3\alpha^3+\lambda_5\alpha^5)}$, up to the identification of (3), which absorbs λ_1 into the linear term; the two parametrisations give the same sum over λ_1 (Lemma 6.1).

2.6. **$\mathbb{Z}/4$-valued quadratic forms.** A $\mathbb{Z}/4$-valued quadratic form on an $\mathbb{F}_2$-space E of dimension n is a map $\mathfrak{q}:E\to\mathbb{Z}/4$ with $\mathfrak{q}(x+y)-\mathfrak{q}(x)-\mathfrak{q}(y)=2b(x,y)$ for a symmetric bilinear form b (the polar form) and $\mathfrak{q}(0)=0$. The radical of b is $\mathcal{R}=\{y: b(y,\cdot)=0\}$; on $\mathcal{R}$ the form $\mathfrak{q}$ is additive with values in $\{0,2\}$. The Gauss sum $\gamma(\mathfrak{q})=\sum_{x\in E}i^{\mathfrak{q}(x)}$ satisfies

$$|\gamma(\mathfrak{q})|^2=\sum_{x,z}i^{\mathfrak{q}(x+z)-\mathfrak{q}(x)}=\sum_z i^{\mathfrak{q}(z)}\sum_x(-1)^{b(x,z)}=2^n\sum_{z\in\mathcal{R}}i^{\mathfrak{q}(z)}=\begin{cases}2^{n+\dim\mathcal{R}} & \text{if }\mathfrak{q}|_{\mathcal{R}}=0,\\ 0 & \text{otherwise,}\end{cases} \tag{8}$$

and when $\gamma(\mathfrak{q})\neq0$, $\gamma(\mathfrak{q})=2^{(n+\dim\mathcal{R})/2}\zeta_8^{\mathrm{Br}(\mathfrak{q})}$ with $\mathrm{Br}(\mathfrak{q})\in\mathbb{Z}/8$ the Brown invariant of the induced nondegenerate form on $E/\mathcal{R}$ [2]. When b is alternating, $\mathfrak{q}$ takes values in $2\mathbb{Z}/4$, $\mathfrak{q}=2Q$ for an $\mathbb{F}_2$-quadratic form Q, and $\mathrm{Br}=4\,\mathrm{Arf}(Q)$; when b is not alternating, $b(x,x)=b(x,w)$ for a unique *characteristic element* $w\in E/\mathcal{R}$. Besides (8) we need the elementary shift identity

$$\sum_{x\in E}i^{\mathfrak{q}(x+c)}=i^{\mathfrak{q}(c)}\sum_{x\in E}i^{\mathfrak{q}(x)+2b(x,c)} \tag{9}$$

and the following observation, which is where the parity of k enters all the sign results of this paper.

Lemma 2.1. *Let $\gamma\in\mathbb{Z}[i]$ with $|\gamma|^2=2^N$. If N is odd then $\gamma/|\gamma|$ is a primitive 8th root of unity; if N is even then $\gamma/|\gamma|\in\{\pm1,\pm i\}$.*

Proof. $\gamma=2^{\lfloor N/2\rfloor}\gamma'$ with $\gamma'\in\mathbb{Z}[i]$ of norm 2 or 1; the elements of norm 2 are the associates of $1+i$, i.e. $\sqrt{2}$ times a primitive 8th root of unity, and those of norm 1 are the units. □

3. The hexad theorem

3.1. **Proof of Theorem 1.1.** Fix $\lambda_3 \in \mathbb{F}_q$, $\lambda_5 \in \mathbb{F}_q^\times$ and drop the index λ. Write $B(u) = \lambda_5^2 u^6 + \lambda_3^2 u^2 + \lambda_5 u + 1$.

Lemma 3.1. $L'(x) = x^4 B(x)^2 + B(x) + 1$.

Proof. $x^4 B(x)^2 = \lambda_5^4 x^{16} + \lambda_3^4 x^8 + \lambda_5^2 x^6 + x^4$ and $B(x) + 1 = \lambda_5^2 x^6 + \lambda_3^2 x^2 + \lambda_5 x$; the terms $\lambda_5^2 x^6$ cancel. □

Lemma 3.2. *B is separable, $B(0) = 1$, and its roots satisfy $e_1 = e_2 = e_3 = 0$, $e_5 = \lambda_5^{-1}$, $e_6 = \lambda_5^{-2} = e_5^2$.*

Proof. $B'(u) = \lambda_5 \neq 0$. The coefficients of u^5, u^4, u^3 vanish, and $e_5 = \lambda_5/\lambda_5^2$, $e_6 = 1/\lambda_5^2$. □

Lemma 3.3. *Let $r \neq s$ be roots of B, $x = r + s$, $p = rs$. Then*

$$B(x) = (\lambda_5 p x)^2 + 1, \qquad \lambda_5^2 p^2 x^3 + x + \lambda_5 p = 0, \qquad L'(x) = x^2\big(\lambda_5^2 p^2 x^3 + x + \lambda_5 p\big)^2 = 0.$$

Proof. Since $B(r) = B(s) = 0$ we have $\lambda_3^2 r^2 + \lambda_5 r = \lambda_5^2 r^6 + 1$ and likewise for s, so

$$B(x) = \lambda_5^2 (r+s)^6 + (\lambda_3^2 r^2 + \lambda_5 r) + (\lambda_3^2 s^2 + \lambda_5 s) + 1 = \lambda_5^2\big((r+s)^6 + r^6 + s^6\big) + 1,$$

and $(r+s)^6 + r^6 + s^6 = (r^2+s^2)(r^4+s^4) + r^6 + s^6 = r^2 s^4 + r^4 s^2 = p^2 x^2$. Next, $u^2 + xu + p$ divides B. Reducing modulo $u^2 + xu + p$: $u^2 \equiv xu + p$, $u^3 \equiv (x^2 + p)u + xp$, $u^6 \equiv (x^2+p)^2(xu+p) + x^2 p^2$, hence

$$B(u) \equiv \big[\lambda_5^2 x (x^2+p)^2 + \lambda_3^2 x + \lambda_5\big] u + \big[\lambda_5^2\big(p(x^2+p)^2 + x^2 p^2\big) + \lambda_3^2 p + 1\big],$$

and both brackets vanish. The first gives $\lambda_3^2 = \lambda_5^2 (x^2+p)^2 + \lambda_5/x$ ($x \neq 0$ as $r \neq s$); substituting into the second, the terms $\lambda_5^2 p (x^2+p)^2$ cancel and there remains $\lambda_5^2 x^2 p^2 + \lambda_5 p/x + 1 = 0$, i.e. $\lambda_5^2 p^2 x^3 + x + \lambda_5 p = 0$. Finally, by Lemma 3.1 and the first formula, with $\beta = B(x) = \lambda_5^2 p^2 x^2 + 1$,

$$\begin{aligned} L'(x) = x^4\beta^2 + \beta + 1 &= x^4(\lambda_5^4 p^4 x^4 + 1) + \lambda_5^2 p^2 x^2 \\ &= x^2\big(\lambda_5^4 p^4 x^6 + x^2 + \lambda_5^2 p^2\big) = x^2\big(\lambda_5^2 p^2 x^3 + x + \lambda_5 p\big)^2. \end{aligned}$$ □

Proof of Theorem 1.1. (a) and (b) are Lemmas 3.1, 3.2 together with the distinctness of the sums: $r_i + r_j = r_i + r_k$ forces $j = k$, and $r_i + r_j = r_k + r_l$ for disjoint pairs gives $r_i + r_j + r_k + r_l = 0 = \sum_t r_t$, hence $r_m + r_n = 0$ for the complementary pair, contradicting separability; $r_i + r_j \neq 0$ for the same reason. (c) By Lemma 3.3 the fifteen sums lie in $\ker L'$, which has exactly 16 elements (L' is separable since its derivative is $\lambda_5 \neq 0$); so they exhaust $\ker L' \setminus 0$. The additional formulas are in the lemma. (d) $L'(r_i) = r_i^4 \cdot 0 + 0 + 1 = 1$, and $L' = (T')^4$ gives $T'(r_i)^4 = 1$, so $T'(r_i) = 1$. The coset $T'^{-1}(1)$ has sixteen elements; the six roots lie in it, and so do the ten sums $r_i + r_j + r_k$ (T' is additive), which are distinct from the roots and from each other ($r_i + r_j + r_k = r_l$ would be a four-term relation, excluded by (b), and $r_i + r_j + r_k = r_i + r_j + r_l$ forces $k = l$; the complementary triple gives the same element because $e_1 = 0$). For x in the coset, $L'(x) = 1$ reads $B(x)\big(x^4 B(x) + 1\big) = 0$, so $B(x) = 0$ or $x^4 B(x) = 1$, and the two cases are the roots and the totals. □

Remark 3.4. The value of B at a nonzero kernel element is $B(r_i + r_j) = (\lambda_5 r_i r_j (r_i + r_j))^2 + 1$, never 0 (a kernel element is not a root, as T' takes the values 0 and 1 on $\mathcal{K}$ and on the roots respectively) and never 1. At a synthematic total t one has $t^4 B(t) = 1$.

Example 3.5 ($q = 8$, $(\lambda_3, \lambda_5) = (0, 1)$)**.** Here $B = u^6 + u + 1$, $L' = x^{16} + x^4 + x$ and $T'(x) = x^4 + x + x^{1/4}$. Let β be a root of $u^6 + u + 1$; it is a primitive element of $\mathbb{F}_{64}$, and the six roots are $\beta, \beta^2, \beta^4, \beta^8, \beta^{16}, \beta^{32}$. Over $\mathbb{F}_8 = \mathbb{F}_2(\beta^9) \subset \mathbb{F}_{64}$ the Frobenius $x \mapsto x^8$ pairs them as (β, β^8), (β^2, β^{16}), (β^4, β^{32}): the class is 2^3, $N_0 = 2$, $\mathcal{K} \subset \mathbb{F}_{q^2} = \mathbb{F}_{64}$. One checks $T'(\beta^{2^j}) = \beta^{2^{j+2}} + \beta^{2^j} + \beta^{2^{j+4}} = 1$ for all j, as Theorem 1.1(d) requires, and the three rational duads $\beta+\beta^8$, $\beta^2+\beta^{16}$, $\beta^4+\beta^{32}$ are the nonzero elements of $\mathcal{K} \cap \mathbb{F}_8$ ($f_1 = 2$). This pair is the single pair of class 2^3 at $q = 8$ (Table 3: $(q-2)/6 = 1$); its W' is $4q^7$, and by Theorem 1.4(d) each of the eight sums $S''(\lambda_1)$ over $\mathbb{F}_{8^{12}} = \mathbb{F}_{2^{36}}$ equals 8^4 times the corresponding sum over $\mathbb{F}_{8^4} = \mathbb{F}_{2^{12}}$, which is $+4q^2$ for every λ_1 (Table 6). Over $\mathbb{F}_2$ the same sextic is irreducible (Example 3.6, class 6): the 8-Frobenius is the cube of the 2-Frobenius, and the cube of a 6-cycle is a product of three transpositions.

Example 3.6 ($q = 2$)**.** For $k = 1$ there are two pairs, $(\lambda_3, \lambda_5) = (0, 1)$ and $(1, 1)$. The sextics are u^6+u+1, which is irreducible over $\mathbb{F}_2$ (class 6, $\mathcal{K} \subset \mathbb{F}_{64}$), and $u^6+u^2+u+1 = (u+1)(u^5+u^4+u^3+u^2+1)$ with an irreducible quintic factor (class $5\,1$, $\mathcal{K} \subset \mathbb{F}_{32}$, $\mathcal{K} \cap \mathbb{F}_{2^{12}} = 0$). The kernels are those of $x^{16} + x^4 + x$ and $x^{16} + x^8 + x^4 + x^2 + x$; the fifteen nonzero elements of the first are the roots of $x^{15} + x^3 + 1$, which factors over $\mathbb{F}_2$ into irreducibles of degrees $3, 6, 6$, the orbit structure of the class 6 in Table 1, and the second gives $x^{15} + x^7 + x^3 + x + 1$, a product of three irreducible quintics (orbits $(5, 5, 5)$). Table 3 is consistent with $q = 2$: all entries vanish except $(q^2+1)/5 = 1$ for the class $5\,1$ and $(q+1)/3 = 1$ for the class 6 in the column $\lambda_3 = 0$.

3.2. The deformation $L_\lambda + \varepsilon x^4$.

Theorem 3.7. *For $\delta \in \bar{\mathbb{F}}$ let $L_{\lambda,\delta^2} = L_\lambda + \delta^2 x^4$ and $B_{\lambda,\delta}(u) = uA_\lambda(u) + \delta = \lambda_5^2 u^6 + \lambda_3^2 u^2 + \lambda_5 u + \delta$. Then*

$$x^4 B_{\lambda,\delta}(x)^2 + B_{\lambda,\delta}(x) + \delta = L_{\lambda,\delta^2}(x),$$

and for $\delta \neq 0$ the kernel of L_{λ,δ^2} is the set of pairwise sums of the six roots of $B_{\lambda,\delta}$; all statements of Theorem 1.1 hold with T'_λ replaced by $T_\lambda + \delta^{1/2}$ and the value 1 by $\delta^{1/2}$. For $\delta = 0$ the roots of $B_{\lambda,0} = uA_\lambda(u)$ are 0 and the five roots of A_λ, and the pairwise-sum description reduces to the quintic frame of §2.3.

Proof. The identity is the computation of Lemma 3.1 with the constant carried along: $x^4 B_{\lambda,\delta}^2 = \lambda_5^4 x^{16} + \lambda_3^4 x^8 + \lambda_5^2 x^6 + \delta^2 x^4$. For $\delta \neq 0$, $B_{\lambda,\delta}$ is separable ($B' = \lambda_5$) with $e_1 = e_2 = 0$, and the proof of Lemma 3.3 goes through verbatim with 1 replaced by δ in the constant term: the relation becomes $\lambda_5^2 p^2 x^3 + \delta x + \lambda_5 p = 0$ and $L_{\lambda,\delta^2}(x) = x^2(\lambda_5^2 p^2 x^3 + \delta x + \lambda_5 p)^2$. For $\delta = 0$ the roots $0, u_0, \dots, u_4$ have pairwise sums u_i and $u_i + u_j$, which are the nonzero elements of $\mathcal{K}_\lambda^0$ by §2.3. □

Remark 3.8 (scaling)**.** Under $x \mapsto sx$ the polynomial $L_{\lambda,\varepsilon}$ becomes $s^{-1} L_{\lambda,\varepsilon}(sx) = \lambda_5^4 s^{15} x^{16} + \lambda_3^4 s^7 x^8 + \varepsilon s^3 x^4 + \lambda_3^2 s x^2 + \lambda_5 x$, so the three-parameter family $(\lambda_3, \lambda_5, \varepsilon)$ modulo scaling is two-dimensional, and $\varepsilon = 1$ is a normalisation when $\varepsilon \neq 0$: the sums W' live on the generic part of the family and the sums W of A_6 on its boundary $\varepsilon = 0$. On the boundary the further scaling symmetry $(\lambda_3, \lambda_5) \mapsto (s^3\lambda_3, s^5\lambda_5)$ survives, which is why W depends only on $j = \lambda_3^5/\lambda_5^3$ while W' does not.

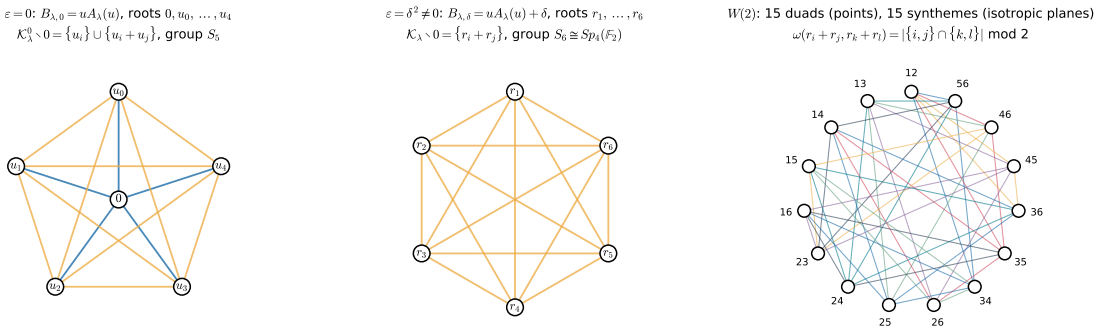


FIGURE 1. Left: the quintic frame ($\varepsilon = 0$); the kernel of L_λ is $\{0\} \cup \{u_i\} \cup \{u_i + u_j\}$ and the Frobenius acts through S_5. Middle: the hexad ($\varepsilon \neq 0$); the kernel of $L_\lambda + \varepsilon x^4$ is $\{0\} \cup \{r_i + r_j\}$ and the Frobenius acts through $S_6 \cong \mathrm{Sp}_4(\mathbb{F}_2)$. Right: the generalised quadrangle $W(2)$ of duads and synthemes, i.e. the points and isotropic lines of $(\mathcal{K}_\lambda, \omega)$.

3.3. **Canonicity.** A hexad with a given duad set is determined up to translation by an element of the kernel; the normalisation $e_2 = 0$ (equivalently $b_4 = 0$ for a monic sextic with $e_1 = 0$) singles out one translate, because translation by c changes the coefficient of u^4 by $\binom{6}{2}c^2 = c^2$. This is how B_λ was found: an exhaustive search over the 168 frames of $\mathcal{K}_\lambda$ for the Frobenius-stable translation classes, followed by this normalisation and a linear fit of the coefficients over $\mathbb{F}_q$ as Laurent polynomials in (λ_3, λ_5) (§7). The fit returned $b_3 = 0$, $b_2 = \lambda_3^2/\lambda_5^2$, $b_1 = \lambda_5^{-1}$, $b_0 = \lambda_5^{-2}$ with a single monomial each, i.e. B_λ/λ_5^2, after which Lemmas 3.1–3.3 were found.

4. THE FROBENIUS CLASSES

4.1. **Proof of Theorem 1.2.** Let σ be the q-Frobenius of $\bar{\mathbb{F}}$. Since $B_\lambda \in \mathbb{F}_q[u]$, σ permutes its roots: $\sigma(r_i) = r_{\pi(i)}$ for a permutation $\pi = \pi_\lambda \in S_6$ whose cycle type is the factorisation type of B_λ over $\mathbb{F}_q$. By Theorem 1.1(c), $\sigma(r_i + r_j) = r_{\pi(i)} + r_{\pi(j)}$: the action on $\mathcal{K}_\lambda \setminus 0$ is the action of π on duads, and it preserves the alternating form (5). The 11 conjugacy classes of S_6 act on the 15 duads with the orbit structures listed in Table 1, which are the eleven patterns of Part II, Table 3 (where they were obtained from the orbits of σ on the roots of L'_λ/x without knowing the hexad). The fixed vectors of σ^m in $\mathcal{K}_\lambda$ are 0 and the duads fixed by π^m, so $f_m := \dim \mathcal{K}_\lambda^{\sigma^m} = \log_2\bigl(1+\#\{\text{duads fixed by } \pi^m\}\bigr)$, which is computed in Table 1. The field $\mathbb{F}_q(\mathcal{K}_\lambda)$ is $\mathbb{F}_{q^{N_0}}$ with N_0 the order of π acting on duads, which is the order of π except for the identity and for the class 2^3, where $\pi = (12)(34)(56)$ has $\pi^2 = 1$ but fixes only the three duads $\{12\}, \{34\}, \{56\}$, so that $N_0 = 2$. That all eleven classes occur for $q \geq 32$ is Table 3 (for $q = 8$ the classes 1^6 and $2\,1^4$ have count 0). □

TABLE 1. The eleven classes. For each cycle type of $\pi \in S_6$: the orbit lengths on the 15 duads (= on $\mathcal{K}_\lambda \setminus 0$), the dimensions $f_m = \dim \mathcal{K}_\lambda^{\sigma^m}$ of the fixed spaces of σ^m for $m = 1, 2, 3, 4, 6$ (these are the radical dimensions of the forms $\mathfrak{q}_\lambda^{(m)}$ of §6), the degree $N_0 = [\mathbb{F}_q(\mathcal{K}_\lambda) : \mathbb{F}_q]$, and the value of W'/q^7 found in Part II (proved here for the class $5\,1$, Theorem 1.4).

class	orbits on duads	f_1	f_2	f_3	f_4	f_6	N_0	$\mathcal{K} \subset \mathbb{F}_{q^{12}}$	W'/q^7
1^6	(1^{15})	4	4	4	4	4	1	yes	4
$2\,1^4$	$(1^7, 2^4)$	3	4	3	4	4	2	yes	0
$2^2\,1^2$	$(1^3, 2^6)$	2	4	2	4	4	2	yes	0
2^3	$(1^3, 2^6)$	2	4	2	4	4	2	yes	4
$3\,1^3$	$(1^3, 3^4)$	2	2	4	2	4	3	yes	4
3^2	(3^5)	0	0	4	0	4	3	yes	4
$3\,2\,1$	$(1, 2, 3, 3, 6)$	1	2	3	2	4	6	yes	0
$4\,1^2$	$(1, 2, 4, 4, 4)$	1	2	1	4	2	4	yes	2
$4\,2$	$(1, 2, 4, 4, 4)$	1	2	1	4	2	4	yes	-2
$5\,1$	$(5, 5, 5)$	0	0	0	0	0	5	no	-1
6	$(3, 6, 6)$	0	0	2	0	4	6	yes	4

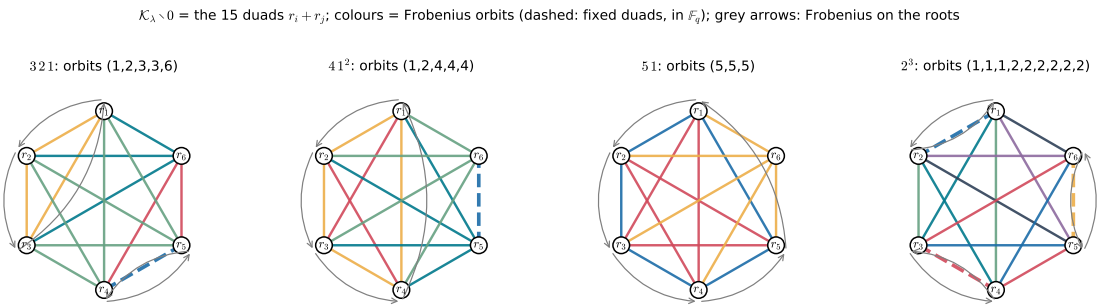


FIGURE 2. The Frobenius on $\mathcal{K}_\lambda \setminus 0$ for four classes. Each panel shows the six roots and the fifteen duads coloured by σ-orbit. The classes $4\,1^2$ and $4\,2$ have the same orbit structure on duads (and so do $2^2 1^2$ and 2^3); the hexad distinguishes them, and so do the Witt sums (Theorem 1.4(d): $W' = +2q^7$ versus $-2q^7$).

Remark 4.1 (why there is no parity obstruction)**.** For the quintic frame of Part II only three of the seven cycle types of S_5 occur, because the radical of the $\mathbb{F}_2$-quadratic form $x \mapsto \mathrm{Tr}_{q/2}(\lambda_5 x^5 + \lambda_3 x^3)$ on $\mathbb{F}_q$ is $\mathcal{K}_\lambda^0 \cap \mathbb{F}_q = (\mathcal{K}_\lambda^0)^\sigma$ and must have dimension $\equiv k \equiv 1 \pmod 2$. For L'_λ the corresponding form on $\mathbb{F}_q$ is $\mathbb{Z}/4$-valued (it is $\mathfrak{q}_\lambda^{(1)}$ of §6) with a non-alternating polar form, and the radical-parity constraint of alternating forms does not apply; indeed Table 1 shows both parities of f_1.

4.2. **Traces on the hexad.** The following lemma, a consequence of Theorem 1.3, is used repeatedly in Section 6. For a root r_i in a π-cycle of length ℓ_i write $c_i = \sum_{t \in \mathrm{cycle}} r_t \in \mathbb{F}_q$ for the cycle sum.

Lemma 4.2. *Let m be a positive integer with $N_0 \mid m$, so that $\mathcal{K}_\lambda \subset \mathbb{F}_{q^m}$.*

(i) *For every root, $\mathrm{Tr}_{q^m/q}(r_i) = (m/\ell_i)\, c_i$. In particular $\mathrm{Tr}_{q^{12}/q}$ vanishes on $\mathcal{K}_\lambda$ unless $N_0 = 4$, in which case $\mathrm{Tr}_{q^{12}/q}(r_i + r_j) = \tau_0 := r_5 + r_6$ (the sum of the two roots outside the 4-cycle) when exactly one of i, j lies in the 4-cycle, and 0 otherwise.*

(ii) *If m is even then $\mathrm{Tr}_{q^m/2}$ vanishes on $\mathcal{K}_\lambda$.*
(iii) *If $N_0 \mid 4$ then $\mathrm{Tr}_{q^4/q}$ vanishes on $\mathcal{K}_\lambda$ unless $N_0 = 4$, where it is as in* (i)*; and $\mathrm{Tr}_{q/2}(\tau_0) = 0$.*

Proof. (i) $\mathrm{Tr}_{q^m/q}(r_i) = \sum_{t=0}^{m-1} \sigma^t(r_i)$ runs m/ℓ_i times over the cycle. For $m = 12$: $12/\ell_i$ is even unless $\ell_i = 4$, and for $\ell_i = 4$ the cycle sum is $e_1 - (r_5 + r_6) = \tau_0$. (ii) $\mathrm{Tr}_{q^m/2}(r_i) = \mathrm{Tr}_{q/2}((m/\ell_i)c_i)$; if m/ℓ_i is even this is 0. Otherwise ℓ_i is even (as m is), so $\ell_i \in \{2, 4, 6\}$: for $\ell_i = 2$, $c_i = r_i + r_i^q$ with $r_i \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$; for $\ell_i = 4$, $c_i = \tau_0 = r_5 + r_6$ where r_5, r_6 are either both in $\mathbb{F}_q$ or conjugate in $\mathbb{F}_{q^2}$; for $\ell_i = 6$, $c_i = e_1 = 0$. In the first two cases $\mathrm{Tr}_{q/2}(c_i) = 0$ by the last assertion of Theorem 1.3. (iii) is (i) with 12 replaced by 4, together with the same trace statement. □

4.3. **Symplectic structures preserved by the Frobenius.** The hexad of B_λ is not the only hexad with duad set $\mathcal{K}_\lambda \setminus 0$ that the Frobenius preserves: by §2.4 the 28 translation classes of hexads correspond to the 28 nondegenerate alternating forms on $\mathcal{K}_\lambda$, and a class is σ-stable iff $\sigma \in \mathrm{Sp}(\omega)$ for the corresponding form. With respect to a stable form ω the element σ has an S_6-class, which depends on ω. The following proposition explains the counts found by `hexad_search.py` (§7) and in particular why the orbit structure of σ on $\mathcal{K}_\lambda \setminus 0$ does not determine the class of B_λ.

Proposition 4.3. *Let $g \in \mathrm{GL}(V) \cong \mathrm{GL}_4(\mathbb{F}_2)$, let C be its conjugacy class in $\mathrm{GL}(V)$, and let t be a conjugacy class of $S_6 \cong \mathrm{Sp}_4(\mathbb{F}_2)$ contained in C (under any of the 28 identifications $S_6 \cong \mathrm{Sp}(V, \omega)$; the image of t in $\mathrm{GL}(V)$ does not depend on ω). Then the number of nondegenerate alternating forms ω on V with $g \in \mathrm{Sp}(V, \omega)$ and g of type t in $\mathrm{Sp}(V, \omega) \cong S_6$ is $28\,|t|/|C|$. The values are given in Table 2.*

Proof. Count pairs (h, ω) with $h \in C$, $h \in \mathrm{Sp}(V, \omega)$ of type t. For fixed ω there are $|t|$ such h, so there are $28|t|$ pairs. The number of ω attached to a given $h \in C$ is constant on C, because $\mathrm{GL}(V)$ permutes the forms transitively and $h \mapsto uhu^{-1}$ carries the forms attached to h to those attached to uhu^{-1} (with the same type). Hence each $h \in C$ has $28|t|/|C|$ forms. □

TABLE 2. The classes of $\mathrm{GL}_4(\mathbb{F}_2)$ met by $S_6 \cong \mathrm{Sp}_4(\mathbb{F}_2)$: for each S_6-class t, the $\mathrm{GL}_4(\mathbb{F}_2)$-class $C \supset t$ described by the Jordan or rational canonical form of its elements and its size $|C| = |\mathrm{GL}_4(\mathbb{F}_2)|/|\mathrm{Cent}(g)|$, and the number $28|t|/|C|$ of g-stable symplectic forms in which g has type t (Proposition 4.3). The last column is the number of Frobenius-stable translation classes of hexads found by `hexad_search.py` for every pair at $q = 8$ and $q = 32$, by canonical class, with the types of σ in those classes (the class $5\,1$ was not searched since $\mathcal{K}_\lambda \not\subset \mathbb{F}_{q^{12}}$).

t	$\lvert t\rvert$	C (canonical form, f_1)	$\lvert C\rvert$	$28\lvert t\rvert/\lvert C\rvert$	found
1^6	1	identity	1	28	28
$2\,1^4$	15	transvection $J_2 \oplus J_1^2$, $f_1 = 3$	105	4	4
$2^2\,1^2$	45	$J_2 \oplus J_2$, $f_1 = 2$	210	6	$8 = 6 + 2$
2^3	15	$J_2 \oplus J_2$, $f_1 = 2$	210	2	
$3\,1^3$	40	$\mathbb{F}_4$-scalar on a plane, $f_1 = 2$	1120	1	1
3^2	40	$\mathbb{F}_4$-scalar on $V \cong \mathbb{F}_4^2$, $f_1 = 0$	112	10	10
$3\,2\,1$	120	order 6, $f_1 = 1$	3360	1	1
$4\,1^2$	90	regular unipotent J_4, $f_1 = 1$	2520	1	$2 = 1 + 1$
$4\,2$	90	regular unipotent J_4, $f_1 = 1$	2520	1	
$5\,1$	144	order 5, $f_1 = 0$	1344	3	–
6	120	order 6, $f_1 = 0$	1680	2	2

Remark 4.4. The class sizes are $|\mathrm{GL}_4(\mathbb{F}_2)| = 20160$ divided by the centraliser orders 20160, 192, 96, 18, 180, 6, 8, 15, 12; e.g. the centraliser of $J_2 \oplus J_2$ is $\mathrm{GL}_2(\mathbb{F}_2[\epsilon]/\epsilon^2)$ of order $6 \cdot 16 = 96$, that of a regular unipotent is the group of units of $\mathbb{F}_2[g]$, of order 8, and the elements of order 3 without fixed vectors are the $\mathbb{F}_4$-scalars of the $112 = |\mathrm{GL}_4(\mathbb{F}_2)|/|\mathrm{GL}_2(\mathbb{F}_4)|$ structures of $\mathbb{F}_4$-space on V. Two consequences: (i) a Frobenius of class 2^21^2 or 2^3 is of type 2^21^2 for six of its eight stable forms and of type 2^3 for the other two; a Frobenius of order 4 is of type $4\,1^2$ for one of its two stable forms and of type $4\,2$ for the other. The value of W' follows the type with respect to the *canonical* form, the one defined by B_λ (Theorem 1.4(d): $+2q^7$ for $4\,1^2$, $-2q^7$ for $4\,2$); so the canonical symplectic structure carries arithmetic information that the $\mathbb{F}_2[\sigma]$-module $\mathcal{K}_\lambda$ alone does not. (ii) A Frobenius of class $5\,1$ preserves three symplectic forms, and is a 5-cycle for all of them; here $\mathcal{K}_\lambda \subset \mathbb{F}_{q^5}$ and the three stable hexads live in $\mathbb{F}_{q^5}$, outside the field $\mathbb{F}_{q^{12}}$ in which the search was carried out.

5. COUNTING

5.1. **The family.** By Lemma 3.2 the map $(\lambda_3, \lambda_5) \mapsto \lambda_5^{-2} B_\lambda$ is a bijection from $\mathbb{F}_q \times \mathbb{F}_q^\times$ onto the set of monic sextics

$$F_{a,b}(u) = u^6 + au^2 + bu + b^2, \qquad a \in \mathbb{F}_q,\ b \in \mathbb{F}_q^\times, \qquad a = \lambda_3^2/\lambda_5^2,\ b = \lambda_5^{-1}, \tag{10}$$

i.e. onto the monic sextics with $e_1 = e_2 = e_3 = 0$ and $e_6 = e_5^2 \neq 0$. The sub-family $a = 0$ ($\lambda_3 = 0$) is $u^6 + bu + b^2$. With $u = bw$, $F_{a,b}(bw) = b^2\big(b^4w^6 + aw^2 + w + 1\big)$, and with $u = 1/v$, $v^6 F_{a,b}(1/v) = b^2v^6 + bv^5 + av^4 + 1$, so that after scaling the family is also $v^6 + v^5 + a'v^4 + c$ with $c = b^4$, $a' = a/b^2$.

5.2. **Proof of Theorem 1.3.** *Rational roots.* A root $u_0 \in \mathbb{F}_q$ of $F_{a,b}$ is nonzero ($F(0) = b^2$), and for given $u_0 \in \mathbb{F}_q^\times$ and $b \in \mathbb{F}_q^\times$ the equation $F_{a,b}(u_0) = 0$ is linear in a with the unique solution $a = (u_0^6 + bu_0 + b^2)/u_0^2 \in \mathbb{F}_q$. Hence $\sum_{(a,b)} N_1 = (q-1)^2$.

Pairs of rational roots. Let $r \neq s$ in $\mathbb{F}_q^\times$ be roots of $F_{a,b}$. Then $s^2F(r) + r^2F(s) = r^2s^2(r^4+s^4) + b\,rs(r+s) + b^2(r+s)^2 = 0$, and dividing by $(r+s)^2$,

$$b^2 + \frac{rs}{r+s}\,b + r^2s^2(r+s)^2 = 0.$$

Writing $b = \frac{rs}{r+s}\,z$ this is $z^2 + z = (r+s)^4$, which has two solutions $z \in \mathbb{F}_q$ iff $\mathrm{Tr}_{q/2}(r+s) = \mathrm{Tr}_{q/2}((r+s)^4) = 0$ and none otherwise; the solutions b are nonzero, and a is then determined. This proves the last assertion of the theorem for rational pairs, and shows that the number of triples $(a,b;r,s)$ with $r \neq s$ rational roots is $2 \cdot \#\{(r,s) \in (\mathbb{F}_q^\times)^2 : r \neq s,\ \mathrm{Tr}(r+s) = 0\}$. For fixed $r \neq 0$ the number of $s \neq 0$ with $\mathrm{Tr}\,s = \mathrm{Tr}\,r$ is $q/2-1$ if $\mathrm{Tr}\,r = 0$ and $q/2$ if $\mathrm{Tr}\,r = 1$, and we must remove $s = r$: the count is $(q/2-1)^2 + (q/2)^2 - (q-1) = q^2/2 - 2q + 2$, and twice this is $(q-2)^2$.

Roots in $\mathbb{F}_{q^2} \setminus \mathbb{F}_q$. Let $u_0 \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$, $\bar{u}_0 = u_0^q$, $s = u_0 + \bar{u}_0 \neq 0$, $n = u_0\bar{u}_0$. For given $b \in \mathbb{F}_q^\times$ the value $a = u_0^4 + b/u_0 + b^2/u_0^2$ lies in $\mathbb{F}_q$ iff it equals its conjugate, i.e. iff $s^4 + bs/n + b^2s^2/n^2 = 0$, i.e. $s^3n^2 + bn + b^2s = 0$. With $n = (b/s^3)m$ this becomes $m^2 + m = s^4$, solvable iff $\mathrm{Tr}_{q/2}(s) = 0$, with solutions $m, m+1$; this is the last assertion of the theorem for conjugate pairs. The pair (s,n) corresponds to a conjugate pair $u_0, \bar{u}_0 \notin \mathbb{F}_q$ iff $u^2 + su + n$ is irreducible, i.e. $\mathrm{Tr}_{q/2}(n/s^2) = \mathrm{Tr}_{q/2}(bm/s^5) = 1$. Hence the number of pairs (b, u_0) is

$$\sum_{\substack{s\neq 0\\ \mathrm{Tr}\,s=0}} \sum_{b\neq 0} 2\Big(\big[\mathrm{Tr}(bm/s^5) = 1\big] + \big[\mathrm{Tr}(b(m+1)/s^5) = 1\big]\Big)$$
$$= \sum_{\substack{s\neq 0\\ \mathrm{Tr}\,s=0}} 2\Big(\frac{q}{2} + \frac{q}{2}\Big) = 2q\Big(\frac{q}{2} - 1\Big) = q(q-2),$$

using that $b \mapsto \mathrm{Tr}(b\gamma)$ takes the value 1 exactly $q/2$ times for $\gamma \neq 0$, that $m, m+1 \neq 0$ because $m^2 + m = s^4 \neq 0$, and that each (s,n) accounts for the two roots $u_0, \bar{u}_0$. Each such pair gives a unique a. □

Corollary 5.1. *Let n_C be the number of $(\lambda_3, \lambda_5) \in \mathbb{F}_q \times \mathbb{F}_q^\times$ whose sextic has class C, and let f_C, $f_C^{(2)}$ be the numbers of fixed points of π and π^2 on $\{1,\dots,6\}$ for $\pi \in C$. Then*

$$\sum_C n_C = q(q-1), \qquad \sum_C n_C f_C = (q-1)^2,$$
$$\sum_C n_C f_C(f_C - 1) = (q-2)^2, \qquad \sum_C n_C\big(f_C^{(2)} - f_C\big) = q(q-2).$$

TABLE 3. Number of $(\lambda_3, \lambda_5) \in \mathbb{F}_q \times \mathbb{F}_q^\times$ with B_λ of a given factorisation type over $\mathbb{F}_q$, k odd: the polynomials of Part II, Table 3 (mixed pairs, $\lambda_3 \neq 0$) and of Part II, Lemma 6.17 and Proposition 6.8 (pure quintic pairs, $\lambda_3 = 0$), verified for all pairs and all $q \in \{8, 32, 128, 512\}$ (`sextic_types.py`). The last column is the proportion of the class in S_6.

class	$\lambda_3 \neq 0$	$\lambda_3 = 0$	$\lvert C\rvert/720$
1^6	$(q-2)(q-8)/720$	0	1/720
$2\,1^4$	$(q-2)(q-8)/48$	0	1/48
$2^2\,1^2$	$q(q-2)/16$	0	1/16
2^3	$(q-2)(q-8)/48$	$(q-2)/6$	1/48
$3\,1^3$	$(q+1)(q-2)/18$	0	1/18
3^2	$(q+1)(q-2)/18$	0	1/18
$3\,2\,1$	$(q+1)(q-2)/6$	0	1/6
$4\,1^2$	$(q-2)(q-4)/8$	$(q-2)/2$	1/8
$4\,2$	$q(q-2)/8$	0	1/8
$5\,1$	$(q^2+1)/5$	0	1/5
6	$(q+1)(q-2)/6$	$(q+1)/3$	1/6

5.3. **The class counts.**

Proposition 5.2. *For $q \in \{8, 32, 128, 512\}$ the numbers of (λ_3, λ_5) with B_λ of each factorisation type are exactly the entries of Table 3. The entries satisfy the four identities of Corollary 5.1 as polynomial identities in q. The column $\lambda_3 = 0$ is a theorem for all odd k.*

Proof. The verification factors the $q(q-1)$ sextics over $\mathbb{F}_q$ by computing $\gcd(B, u^{q^m} - u)$ for $m \leq 6$ ($q = 512$: 261 632 sextics). The identities are checked by expanding; e.g. $\sum n_C f_C = 6\frac{(q-2)(q-8)}{720} + 4\frac{(q-2)(q-8)}{48} + 2\frac{q(q-2)}{16} + 3\frac{(q+1)(q-2)}{18} + \frac{(q+1)(q-2)}{6} + 2\frac{(q-2)(q-4)}{8} + \frac{q^2+1}{5} + 2\cdot\frac{q-2}{2} = (q-1)^2$.

For $\lambda_3 = 0$, Part II, Lemma 6.17, shows that $\mathcal{K}_{(0,\lambda_5)}$ is obtained from the quintic frame of the Gold quintic $v^5 + v + \lambda_5^2$ by extracting cube roots, with orbit structure on $\mathcal{K} \setminus 0$ equal to $(1, 2, 4, 4, 4)$, $(1^3, 2^6)$, $(3, 6, 6)$ for the quintic types $(1, 4)$, $(1, 1, 1, 2)$, $(2, 3)$, whose numbers are $(q-2)/2$, $(q-2)/6$, $(q+1)/3$ by Part II, Proposition 6.8. By Table 1 the sextic classes with these orbit structures are $\{4\,1^2, 4\,2\}$, $\{2^2 1^2, 2^3\}$ and $\{6\}$ respectively. The first moment decides between the alternatives: the number of rational roots of $u^6 + bu + b^2$ summed over $b \in \mathbb{F}_q^\times$ is, by the computation above with $a = 0$ (for $u_0 \in \mathbb{F}_q^\times$ the equation $b^2 + u_0 b + u_0^6 = 0$, i.e. $z^2 + z = u_0^4$ with $b = u_0 z$, has two solutions iff $\operatorname{Tr} u_0 = 0$), equal to $2(q/2 - 1) = q - 2$. The classes $4\,1^2$ and $2^2 1^2$ have two rational roots each, $4\,2$ and 2^3 none; so $q - 2 = 2\alpha\frac{q-2}{2} + 2\beta\frac{q-2}{6}$ with $\alpha, \beta \in \{0, 1\}$ indicating the choices $4\,1^2$ and $2^2 1^2$, whose only solution is $\alpha = 1$, $\beta = 0$. □

Conjecture 5.3. *Table 3 holds for all odd k.*

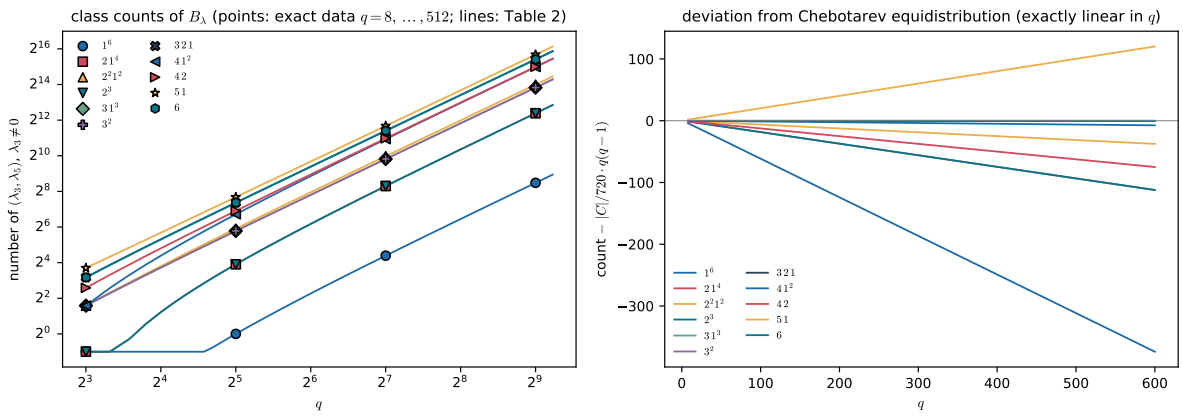


FIGURE 3. Left: the class counts of B_λ for $\lambda_3 \neq 0$ (points: exact data for $q = 8, 32, 128, 512$; lines: the polynomials of Table 3; counts equal to 0 are drawn at 2^{-1}). Right: the deviation of each count from the Chebotarev proportion $|C|/720 \cdot q(q-1)$ is exactly linear in q.

Remark 5.4. The leading coefficients are the class proportions of S_6: the family is equidistributed in the sense of Chebotarev, as expected since its geometric monodromy group is S_6 (S_6 is the only subgroup of S_6 containing a 5-cycle, a 6-cycle and a transposition). What is remarkable is the absence of any error term: the counts are polynomials with the same value for every odd k. For the quintic v^5+v+c this exactness was proved in Part II by counting roots in $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$ (two linear relations suffice for three classes). For the sextic the analogous relations are the four of Corollary 5.1, which do not determine eleven unknowns; the higher moments (N_m for $m \leq 6$ and the second moments over $\mathbb{F}_{q^2}, \mathbb{F}_{q^3}$) are counts of points on explicit curves over $\mathbb{F}_q$, e.g. N_3 is the number of $(w, a') \in \mathbb{F}_{q^3} \times \mathbb{F}_q$ with $w^4(w^2+w+a') \in \mathbb{F}_q$ in the normalisation $v^6+v^5+a'v^4+c$, and the exactness of Table 3 says that all these curves have polynomial point counts. We leave this as Open Problem **Open Problem 1.**.

6. The Witt sums

6.1. Two parametrisations of the order-four characters.

Lemma 6.1. *Let $\tau = \mathrm{Tr}_{q^{12}/q}$ and let $S'(\lambda)$, $S''(\lambda)$ be as in (1), (3). For $x \in \mathbb{F}_{q^{12}}$,*

$$\mathsf{T}_{12k}(x) = \mathrm{Tr}_{q/2}(\tau x) + 2\big(\mathrm{Tr}_{q/2}\, e_2(x) + e_2^{\mathrm{abs},\mathbb{F}_q}(\tau x)\big),$$

where $e_2(x)$ is relative to $\mathbb{F}_q$ (twelve conjugates) and $e_2^{\mathrm{abs},\mathbb{F}_q}(y) = \sum_{0\leq i<j<k} y^{2^i+2^j}$ for $y \in \mathbb{F}_q$. Consequently $S'(\lambda)$ and $S''(\lambda)$ have the same restriction to $\{\tau x = 0\}$, and

$$\sum_{\lambda_1} S'(\lambda) = \sum_{\lambda_1} S''(\lambda) = q \sum_{\tau x = 0} (-1)^{\mathrm{Tr}(\lambda_5 x^5 + \lambda_3 x^3) + \mathrm{Tr}_{q/2}\, e_2(x)} = W'(\lambda_3, \lambda_5),$$

but $S'(\lambda) \neq S''(\lambda)$ in general.

Proof. Transitivity of the Witt trace: $\mathrm{Tr}_{W_2(\mathbb{F}_{q^{12}})/\mathbb{Z}/4}[x] = \mathrm{Tr}_{W_2(\mathbb{F}_q)/\mathbb{Z}/4}\big(\mathrm{Tr}_{W_2(\mathbb{F}_{q^{12}})/W_2(\mathbb{F}_q)}[x]\big)$, and the inner trace is the Witt sum of the twelve Teichmüller lifts of the conjugates, $(\tau x, e_2(x))$; then $(y_0, y_1) = [y_0] + (0, y_1)$ and $\mathrm{Tr}_{W_2(\mathbb{F}_q)/\mathbb{Z}/4}(0, y_1) = 2\,\mathrm{Tr}_{q/2}\, y_1$. The sum over λ_1 of $(-1)^{\mathrm{Tr}(\lambda_1 x)}$ is $q[\tau x = 0]$ in both parametrisations. □

Thus W' does not depend on which of the two families of order-4 characters is used, but only S'' has an L-function: the sheaf $\mathcal{L}_{\psi_4}(x, f_\lambda(x))$ on $\mathbb{A}^1_{\mathbb{F}_q}$, $f_\lambda = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$,

for the character $\psi_4(a_0+2a_1)=i^{a_0+2a_1}$ of $\mathbb{Z}/4=W_2(\mathbb{F}_2)$, has trace function $x\mapsto i^{\mathsf{T}_{mk}(x)+2\operatorname{Tr}_{q^m/2}f_\lambda(x)}$ at $x\in\mathbb{F}_{q^m}$ (the Witt vector $(x,f_\lambda(x))=[x]+Vf_\lambda(x)$), is lisse of rank one and pure of weight 0 on $\mathbb{A}^1$, and has Swan conductor $\max(2\deg x,\deg f_\lambda)=5$ at infinity when $\lambda_5\neq 0$ [3, 10]. By the Grothendieck–Ogg–Shafarevich formula $\chi_c(\mathbb{A}^1_{\overline{\mathbb{F}}},\mathcal{L})=1-5=-4$, and $H^0_c=H^2_c=0$ as the sheaf is nontrivial and lisse on an affine curve; so H^1_c has rank 4 with Frobenius eigenvalues $\omega_1,\dots,\omega_4$, and by Deligne [7] $|\omega_j|\le\sqrt{q}$ for every complex embedding. Writing

$$s_m(\lambda):=\sum_{x\in\mathbb{F}_{q^m}} i^{\mathfrak{q}_\lambda^{(m)}(x)},\qquad \mathfrak{q}_\lambda^{(m)}(x):=\mathsf{T}_{mk}(x)+2\operatorname{Tr}_{q^m/2}f_\lambda(x), \tag{11}$$

the Lefschetz trace formula gives

$$s_m(\lambda)=-\sum_{j=1}^{4}\omega_j^m\quad(m\ge 1),\qquad S''(\lambda)=s_{12}(\lambda). \tag{12}$$

6.2. The $\mathbb{Z}/4$-forms and their radicals.

Proposition 6.2. *For every $m\ge 1$, $\mathfrak{q}_\lambda^{(m)}$ is a $\mathbb{Z}/4$-valued quadratic form on $\mathbb{F}_{q^m}$ with polar form $b_\lambda^{(m)}(x,y)=\operatorname{Tr}_{q^m/2}\big(x\,T'_\lambda(y)\big)$, radical $\mathcal{R}_\lambda^{(m)}=\mathcal{K}_\lambda\cap\mathbb{F}_{q^m}=\mathcal{K}_\lambda^{\sigma^m}$ of dimension f_m (Table 1), and characteristic element any element of $T'^{-1}_\lambda(1)\cap\mathbb{F}_{q^m}$ when this set is nonempty. Consequently*

$$s_m(\lambda)\in\{0\}\cup 2^{(mk+f_m)/2}\mu_8,\qquad s_m(\lambda)\neq 0\iff \mathfrak{q}_\lambda^{(m)}|_{\mathcal{K}_\lambda\cap\mathbb{F}_{q^m}}=0.$$

On the radical, $\mathfrak{q}_\lambda^{(m)}(x)=\mathsf{T}_{mk}(x)+2\operatorname{Tr}_{q^m/2}(\lambda_5x^5+\lambda_3x^3)+2\operatorname{Tr}_{q/2}\big(\lambda_1\operatorname{Tr}_{q^m/q}x\big)$ is additive and σ-invariant.

Proof. By (7) the polar form of T_{mk} is $2\operatorname{Tr}_{q^m/2}(xy)$; that of $2\operatorname{Tr}_{q^m/2}f_\lambda$ is $2\operatorname{Tr}_{q^m/2}(xT_\lambda(y))$ (the polar form of the $\mathbb{F}_2$-form $\operatorname{Tr}(\lambda_5x^5+\lambda_3x^3)$ is $\operatorname{Tr}(y(D_\lambda+D_\lambda^*)(x))$, and the linear term has none). Adding, $b_\lambda^{(m)}(x,y)=\operatorname{Tr}_{q^m/2}(x(T_\lambda+1)(y))$; note that T'_λ maps $\mathbb{F}_{q^m}$ to itself. The radical is $\ker T'_\lambda\cap\mathbb{F}_{q^m}$ by nondegeneracy of the trace form. Next $b(x,x)=\operatorname{Tr}_{q^m/2}(xT'_\lambda(x))=\operatorname{Tr}_{q^m/2}(x^2)+\operatorname{Tr}_{q^m/2}(\lambda_5x^5+\lambda_3x^3)+\operatorname{Tr}_{q^m/2}\big(x(\lambda_3x)^{1/2}+x(\lambda_5x)^{1/4}\big)=\operatorname{Tr}_{q^m/2}(x)$, because $\operatorname{Tr}(x(\lambda_3x)^{1/2})=\operatorname{Tr}((\lambda_3x^3)^{1/2})=\operatorname{Tr}(\lambda_3x^3)$ and similarly for the quintic term; and $\operatorname{Tr}(x)=\operatorname{Tr}(xT'_\lambda(w))$ for all x iff $T'_\lambda(w)=1$. The rest is (8) and $\dim\mathcal{R}^{(m)}=f_m$ (Theorem 1.2). □

6.3. The multiplicity formula along towers.

Lemma 6.3. *Let $m\mid M$ and $x\in\mathbb{F}_{q^m}\subset\mathbb{F}_{q^M}$. Then, in $\mathbb{Z}/4$,*

$$\mathfrak{q}_\lambda^{(M)}(x)=\frac{M}{m}\cdot\mathfrak{q}_\lambda^{(m)}(x)=\begin{cases}\mathfrak{q}_\lambda^{(m)}(x) & M/m\equiv 1 \pmod 4,\\ 2\operatorname{Tr}_{q^m/2}(x) & M/m\equiv 2 \pmod 4,\\ \mathfrak{q}_\lambda^{(m)}(x)+2\operatorname{Tr}_{q^m/2}(x) & M/m\equiv 3 \pmod 4,\\ 0 & M/m\equiv 0 \pmod 4.\end{cases}$$

In particular, for $M=12$: $\mathfrak{q}_\lambda^{(12)}$ vanishes identically on $\mathbb{F}_q$ and on $\mathbb{F}_{q^3}$, equals $2\operatorname{Tr}_{q^2/2}$ on $\mathbb{F}_{q^2}$ and $2\operatorname{Tr}_{q^6/2}$ on $\mathbb{F}_{q^6}$, and equals $\mathfrak{q}_\lambda^{(4)}+2\operatorname{Tr}_{q^4/2}$ on $\mathbb{F}_{q^4}$.

Proof. $\operatorname{Tr}_{W_2(\mathbb{F}_{q^M})/W_2(\mathbb{F}_{q^m})}[x]$ is the sum of M/m copies of $[x]$, i.e. $(M/m)\cdot[x]$ with $M/m\in\mathbb{Z}/4=W_2(\mathbb{F}_2)$ acting through the ring structure; the trace $W_2(\mathbb{F}_{q^m})\to\mathbb{Z}/4$ is $\mathbb{Z}/4$-linear, so $\mathsf{T}_{Mk}(x)=(M/m)\mathsf{T}_{mk}(x)$. Also $2\operatorname{Tr}_{q^M/2}f_\lambda(x)=2(M/m)\operatorname{Tr}_{q^m/2}f_\lambda(x)$. The case distinction uses $2v=2(v \bmod 2)$ and $3v=v+2(v \bmod 2)$ in $\mathbb{Z}/4$ together with $\mathfrak{q}_\lambda^{(m)}(x)\equiv\operatorname{Tr}_{q^m/2}(x) \pmod 2$. □

6.4. **Integrality and the shift calculus.**

Lemma 6.4 (shifts). *Let $c \in \mathbb{F}_{q^m}$ with $T'_\lambda(c) = a \in \mathbb{F}_q$. Then, with λ_3, λ_5 fixed,*

$$s_m(\lambda_1) = i^{\mathfrak{q}_\lambda^{(m)}(c)}\, s_m(\lambda_1 + a).$$

For $m = 12$ and $c \in \mathbb{F}_q$ the phase is trivial, so S'' is constant on cosets of $T'_\lambda(\mathbb{F}_q)$ in $\mathbb{F}_q$.

Proof. (9) with $2b^{(m)}(x,c) = 2\operatorname{Tr}_{q^m/2}(xT'_\lambda(c)) = 2\operatorname{Tr}_{q^m/2}(ax)$, which is the change $\lambda_1 \mapsto \lambda_1 + a$ in $2\operatorname{Tr} f_\lambda$. For $c \in \mathbb{F}_q$, $\mathfrak{q}_\lambda^{(12)}(c) = 0$ by Lemma 6.3. □

Proposition 6.5 (integrality). *For all λ, with λ_3, λ_5 fixed: $\overline{S''(\lambda_1)} = S''(\lambda_1{+}1) = S''(\lambda_1)$. Hence $S''(\lambda) \in \mathbb{Z}$, and more precisely $S''(\lambda) \in \{0, \pm 4q^6\}$ if $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{12}}$ and $S''(\lambda) = \pm q^6$ otherwise.*

Proof. Complex conjugation replaces $\mathfrak{q}$ by $-\mathfrak{q} = \mathfrak{q} + 2(\mathfrak{q} \bmod 2) = \mathfrak{q} + 2\operatorname{Tr}_{q^{12}/2}(x)$, which is $\mathfrak{q}_{\lambda_1+1}$: this is the first equality. For the second, pick a root r of B_λ in $\mathbb{F}_{q^{12}}$ with $\mathfrak{q}_\lambda^{(12)}(r) = 0$; such a root exists: if r lies in a π-cycle of length $\ell \in \{1,3\}$ then $r \in \mathbb{F}_{q^\ell}$ and $\mathfrak{q}^{(12)}(r) = (12/\ell)\mathfrak{q}^{(\ell)}(r) = 0$ by Lemma 6.3; if $\ell \in \{2,6\}$ then $\mathfrak{q}^{(12)}(r) = 2\operatorname{Tr}_{q^\ell/2}(r) = 2\operatorname{Tr}_{q/2}(c)$ with $c = \operatorname{Tr}_{q^\ell/q}(r)$ the cycle sum, which has trace 0 by Theorem 1.3 ($\ell = 2$) or vanishes because $c = e_1 = 0$ ($\ell = 6$); and every class has a root with $\ell \in \{1,2,3,6\}$ (in the classes $4\,1^2$, $4\,2$ take r_5, in the class $5\,1$ take the rational root). Since $T'_\lambda(r) = 1$, Lemma 6.4 gives $S''(\lambda_1) = i^0 S''(\lambda_1 + 1)$. Thus $S''(\lambda)$ is real, and by Proposition 6.2 it is 0 or $2^{(12k+f_{12})/2}$ times an 8th root of unity, with $f_{12} = 4$ if $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{12}}$ and $f_{12} = 0$ otherwise; a real element of μ_8 is ± 1. In the class $5\,1$ the radical is 0, so $S'' \neq 0$. □

6.5. **The eigenvalues are roots of unity.**

Theorem 6.6. *Let $N_0 = [\mathbb{F}_q(\mathcal{K}_\lambda) : \mathbb{F}_q]$. Then $s_{4N_0}(\lambda) \neq 0$, $|s_{4N_0}(\lambda)| = 4q^{2N_0}$, and*

$$\omega_1^{4N_0} = \omega_2^{4N_0} = \omega_3^{4N_0} = \omega_4^{4N_0} = -\tfrac{1}{4}s_{4N_0}(\lambda) = q^{2N_0}\nu, \qquad \nu \in \mu_8.$$

In particular $|\omega_j| = \sqrt{q}$ and $\zeta_j := \omega_j/\sqrt{q}$ satisfies $\zeta_j^{32N_0} = 1$.

Proof. $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{N_0}} \subset \mathbb{F}_{q^{4N_0}}$, so the radical of $\mathfrak{q}_\lambda^{(4N_0)}$ is all of $\mathcal{K}_\lambda$ (dimension 4), and $\mathfrak{q}_\lambda^{(4N_0)}$ vanishes on $\mathbb{F}_{q^{N_0}}$ by Lemma 6.3 ($M/m = 4$). Hence $s_{4N_0} = 2^{(4N_0k+4)/2}\zeta_8^b = 4q^{2N_0}\zeta_8^b$ by (8). Now $-s_{4N_0} = \sum_j \omega_j^{4N_0}$ is a sum of four complex numbers of modulus $\le q^{2N_0}$ (Deligne) whose modulus is $4q^{2N_0}$; so all four are equal to $-s_{4N_0}/4$ and have modulus q^{2N_0}. □

Corollary 6.7. *The four Gauss sums $s_1, \dots, s_4$ over $\mathbb{F}_q, \dots, \mathbb{F}_{q^4}$ determine the ω_j (Newton's identities) and hence $S''(\lambda) = -\sum_j \omega_j^{12}$. If $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{12}}$ then $S''(\lambda) \neq 0$ iff $\omega_1^{12} = \cdots = \omega_4^{12} = -S''(\lambda)/4 = \mp q^6$, i.e. iff $\zeta_j^{12} = \mp 1$ for all j, where $S''(\lambda) = \pm 4q^6$; in that case the ζ_j are 24th roots of unity.*

Proof. The first statement is (12). If $S'' = \pm 4q^6$ then $|\sum \omega_j^{12}| = 4q^6$ with $|\omega_j^{12}| = q^6$ forces equality of the four terms; conversely if all ω_j^{12} are equal to $\mp q^6$ then $S'' = \pm 4q^6$. □

Remark 6.8. Thus, when $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{12}}$, the sign of a nonzero $S''(\lambda)$ is the statement that the geometric Frobenius over $\mathbb{F}_{q^{12}}$ acts on H_c^1 as the scalar $-q^6$ ($S'' = +4q^6$) or $+q^6$ ($S'' = -4q^6$), and $S''(\lambda) = 0$ means that $\omega_j^{12} = q^6$ for two j and $-q^6$ for the other two. This is the Witt-sheaf analogue of the supersingularity statement $\mathrm{Frob}^N = \pm q^{N/2}$ for the Artin–Schreier curves of Part II, and all three cases occur (Table 5, Figure 4).

6.6. **The class of 5-cycles.**

Theorem 6.9. *Suppose B_λ has an irreducible quintic factor over $\mathbb{F}_q$ (k odd). Then:*

(i) *the normalised eigenvalues $\zeta_j = \omega_j/\sqrt{q}$ are distinct and ζ_j^5 is the same for all j;*
(ii) *$s_1(\lambda) = \eta\sqrt{q}$ with η a primitive 8th root of unity, and $\{\zeta_1,\dots,\zeta_4\} = \{\eta\zeta_5^c : c = 1,2,3,4\}$;*
(iii) *$\prod_j(1-\omega_j T) = \dfrac{1-\eta^5 q^{5/2}T^5}{1-\eta q^{1/2}T} = 1 + s_1T + s_1^2T^2 + s_1^3T^3 + s_1^4T^4$, and $s_m(\lambda) = s_1(\lambda)^m$ for all m with $5\nmid m$, $s_m(\lambda) = -4s_1(\lambda)^m$ for $5 \mid m$;*
(iv) *$S''(\lambda_1,\lambda_3,\lambda_5) = q^6\eta^4 = -q^6$ for every $\lambda_1 \in \mathbb{F}_q$, and $W'(\lambda_3,\lambda_5) = -q^7$.*

Proof. Here $N_0 = 5$, the orbits of σ on the duads are $(5,5,5)$, and π^m is again a 5-cycle for $5\nmid m$; so $f_m = 0$ for $5\nmid m$ and $f_m = 4$ for $5 \mid m$ (Theorem 1.2). By Proposition 6.2, $|s_m| = q^{m/2}$ for $5\nmid m$ (a Gauss sum with trivial radical is nonzero) and $|s_m| \in \{0, 4q^{m/2}\}$ for $5 \mid m$. Put $p_m = \sum_j \zeta_j^m = -s_m/q^{m/2}$. By Theorem 6.6 the ζ_j are roots of unity, of order dividing some N which we may take divisible by 5. Averaging over a period,

$$\frac{1}{N}\sum_{m=1}^{N}|p_m|^2 = \sum_{j,l}\frac{1}{N}\sum_{m=1}^{N}(\zeta_j\bar{\zeta_l})^m = \#\{(j,l) : \zeta_j = \zeta_l\} \;\geq\; 4,$$

while $|p_m|^2 = 1$ for the $4N/5$ indices with $5\nmid m$ and $|p_m|^2 \leq 16$ for the $N/5$ others, so the average is $\leq \frac{4}{5} + \frac{16}{5} = 4$. Hence equality holds throughout: the ζ_j are distinct, and $|p_m| = 4$ for every $m \equiv 0 \pmod 5$, in particular $p_5 = 4\nu$ with $\zeta_j^5 = \nu$ for all j. This is (i). Write $\zeta_j = \rho\zeta_5^{a_j}$ with $\rho^5 = \nu$; the a_j are four distinct residues mod 5, and if a_0 is the missing one and $\eta := \rho\zeta_5^{a_0}$, then $\{\zeta_j\} = \{\eta\zeta_5^c : c \neq 0\}$ and $p_1 = \eta\sum_{c\neq 0}\zeta_5^c = -\eta$, so $s_1 = -\sqrt{q}\,p_1 = \eta\sqrt{q}$. Now $s_1 \in \mathbb{Z}[i]$ with $|s_1|^2 = q = 2^k$, k odd, so η is a primitive 8th root of unity by Lemma 2.1: this is (ii). The L-polynomial is $\prod_{c\neq 0}(1-\eta\zeta_5^c\sqrt{q}T) = (1-\eta^5q^{5/2}T^5)/(1-\eta\sqrt{q}T)$, whose expansion is $\sum_{t=0}^{4}(\eta\sqrt{q})^tT^t$; and $s_m = -q^{m/2}\eta^m\sum_{c\neq 0}\zeta_5^{mc}$ equals $q^{m/2}\eta^m = s_1^m$ for $5\nmid m$ and $-4s_1^m$ for $5 \mid m$: this is (iii). Finally $S'' = s_{12} = s_1^{12} = q^6\eta^{12} = q^6\eta^4 = -q^6$, for every λ_1, since the argument used only the class of (λ_3,λ_5); and $W' = \sum_{\lambda_1} S'' = -q^7$. □

Remark 6.10. The theorem does not use the values of Part II; it is a complete, independent evaluation of W' in the class 5 1. The mechanism is worth stating in words: the Frobenius has no fixed vector in $\mathcal{K}_\lambda$ over $\mathbb{F}_{q^m}$ for $5\nmid m$, so the Witt sums over those fields have the smallest possible modulus $q^{m/2}$; this rigidity forces the four eigenvalues to be the 5th-root-of-unity translates of a single number $\eta\sqrt{q}$, and the parity of k forces η to be primitive. For even k the same argument gives $\eta \in \{\pm 1, \pm i\}$ and $S'' = +q^6$, in line with the sign $(-1)^{k+1}$ in front of q^6 in the conjectural formula $X = (-1)^{k+1}q^6 + q^5$ of Part II.

Example 6.11 ($q = 8$, class 5 1)**.** For 39 of the 104 triples $(\lambda_3,\lambda_5,\lambda_1)$ of this class at $q = 8$ one finds $s_1 = 2 + 2i = \sqrt{8}\,\zeta_8$ (Table 6, $b = 1$), hence $s_2 = s_1^2 = 8i$, $s_3 = s_1^3 = -16 + 16i$, $s_4 = s_1^4 = -64$, and the L-polynomial of the Witt sheaf is

$$\prod_{j=1}^{4}(1-\omega_jT) = 1 + (2+2i)T + 8iT^2 + (-16+16i)T^3 - 64T^4 = \frac{1 + (128+128i)T^5}{1-(2+2i)T},$$

with $S'' = s_1^{12} = (8i)^6 = -8^6 = -q^6$. The other values $b = 3,5,7$ give the Galois conjugates $s_1 = -2+2i, -2-2i, 2-2i$ with the same s_4 and the same S''.

6.7. **The classes with $N_0 \mid 4$.**

Theorem 6.12. *Suppose $\mathcal{K}_\lambda \subset \mathbb{F}_{q^4}$, i.e. B_λ has class 1^6, $2\,1^4$, $2^2 1^2$, 2^3, $4\,1^2$ or $4\,2$. Then*

$$S''(\lambda) = q^4\, s_4(\lambda) \qquad \text{and} \qquad s_4(\lambda) \in \{0, \pm 4q^2\}.$$

Moreover: if $N_0 \mid 2$ then $s_4(\lambda) \neq 0$ for every λ_1; if $N_0 = 4$ then $\{\lambda_1 : s_4(\lambda) \neq 0\}$ is either empty or the affine hyperplane $\{\lambda_1 : \mathrm{Tr}_{q/2}(\lambda_1 \tau_0) = \kappa_0\}$ for an explicit $\kappa_0 \in \mathbb{F}_2$, where $\tau_0 = r_5 + r_6$ is the sum of the roots outside the 4-cycle.

Proof. By Lemma 6.3, $\mathfrak{q}_\lambda^{(12)} = \mathfrak{q}_\lambda^{(4)} + 2\,\mathrm{Tr}_{q^4/2}$ on $\mathbb{F}_{q^4}$, and $\mathrm{Tr}_{q^4/2}$ vanishes on $\mathcal{K}_\lambda$ by Lemma 4.2(ii). Hence $\mathfrak{q}_\lambda^{(12)}|_{\mathcal{K}_\lambda} = \mathfrak{q}_\lambda^{(4)}|_{\mathcal{K}_\lambda}$, and since $\mathcal{K}_\lambda$ is the radical of both forms, $S''(\lambda) \neq 0 \iff s_4(\lambda) \neq 0$ (Proposition 6.2). If both vanish the identity holds. Otherwise $s_4 = 4q^2\zeta_8^b$ with b even (Lemma 2.1, $|s_4|^2 = 2^{4k+4}$), so $\nu := -s_4/4q^2 \in \{\pm 1, \pm i\}$, and as in Theorem 6.6 all ω_j^4 equal $q^2\nu$; then $S'' = -\sum_j \omega_j^{12} = -4q^6\nu^3$. By Proposition 6.5 this is real, so $\nu^3 \in \mathbb{R}$, $\nu = \pm 1$, $\nu^3 = \nu$ and $S'' = -4q^6\nu = q^4 s_4$. In particular $s_4 \in \{0, \pm 4q^2\}$.

For the vanishing: by Proposition 6.2 the dependence of $\mathfrak{q}_\lambda^{(4)}|_{\mathcal{K}_\lambda}$ on λ_1 is through $2\,\mathrm{Tr}_{q/2}(\lambda_1 \mathrm{Tr}_{q^4/q}\, x)$. If $N_0 \mid 2$ then $\mathfrak{q}_\lambda^{(4)} = 2\mathfrak{q}_\lambda^{(2)} = 2\,\mathrm{Tr}_{q^2/2}$ on $\mathbb{F}_{q^2} \supset \mathcal{K}_\lambda$ (Lemma 6.3), which vanishes on $\mathcal{K}_\lambda$ by Lemma 4.2(ii); so $s_4 \neq 0$ for all λ_1. If $N_0 = 4$, let (1234) be the 4-cycle. $\mathcal{K}_\lambda$ is spanned by $r_1 + r_2$, its σ-images, and $r_1 + r_5$; $\mathrm{Tr}_{q^4/q}$ vanishes on the duads inside the 4-cycle and on $r_5 + r_6$ and equals $\tau_0 \neq 0$ on $r_1 + r_5$ (Lemma 4.2(iii)); as $\mathfrak{q}_\lambda^{(4)}$ is additive and σ-invariant on $\mathcal{K}_\lambda$, its vanishing is equivalent to the vanishing of the λ_1-independent values at $r_1 + r_2$ and $r_5 + r_6$ together with $\mathfrak{q}^{(4)}_{\lambda,\lambda_1=0}(r_1 + r_5) + 2\,\mathrm{Tr}_{q/2}(\lambda_1\tau_0) = 0$, i.e. $\mathrm{Tr}_{q/2}(\lambda_1\tau_0) = \kappa_0 := \frac{1}{2}\mathfrak{q}^{(4)}_{\lambda,\lambda_1=0}(r_1 + r_5)$. □

Corollary 6.13. *In the classes $4\,1^2$ and $4\,2$, all nonzero $S''(\lambda_1, \lambda_3, \lambda_5)$ have the same sign, namely that of s_4, and $W'(\lambda_3, \lambda_5) = \frac{q}{2} \cdot S''(\lambda_1^*, \lambda_3, \lambda_5) \in \{0, \pm 2q^7\}$ for any λ_1^* with $S'' \neq 0$. In the classes 1^6, $2\,1^4$, $2^2 1^2$, 2^3, every $S''(\lambda_1, \lambda_3, \lambda_5)$ is $\pm 4q^6$ and $W' = 4q^6 \sum_{\lambda_1} \mathrm{sgn}\, s_4(\lambda_1, \lambda_3, \lambda_5)$.*

Proof. Only the constancy of the sign on the hyperplane needs proof. $T'_\lambda(\mathbb{F}_q)$ has codimension $f_1 = 1$ in $\mathbb{F}_q$ and S'' is constant on its cosets (Lemma 6.4); the hyperplane $\{\mathrm{Tr}(\lambda_1\tau_0) = \kappa_0\}$ is a coset of $\ker \mathrm{Tr}_{q/2}(\cdot\,\tau_0)$, which contains $T'_\lambda(\mathbb{F}_q)$, because

$$\mathrm{Tr}_{q/2}\big(T'_\lambda(c)\,\tau_0\big) = \mathrm{Tr}_{q/2}\big(c\,T'_\lambda(\tau_0)\big) = 0$$

(T'_λ is self-adjoint and $\tau_0 \in \mathcal{K}_\lambda \cap \mathbb{F}_q$); both have codimension one, so the hyperplane is a single coset of $T'_\lambda(\mathbb{F}_q)$. □

6.8. **The classes with $N_0 \in \{3, 6\}$.**

Theorem 6.14. *Suppose $N_0 \in \{3, 6\}$, i.e. B_λ has class $3\,1^3$, 3^2, $3\,2\,1$ or 6. Then:*

(i) *$S''(\lambda_1, \lambda_3, \lambda_5) = \pm 4q^6$ for every λ_1;*
(ii) *if $N_0 = 3$ and $s_3(\lambda) \neq 0$ then $S''(\lambda) = +4q^6$;*
(iii) *if $s_6(\lambda) \neq 0$ then $\nu_6 := -s_6(\lambda)/4q^3 \in \{\pm 1, \pm i\}$ and $S''(\lambda) = -4q^6\nu_6^2$, i.e. $S'' = +4q^6$ iff $\nu_6 = \pm i$;*
(iv) *in the classes 3^2 and 6 all q values $S''(\lambda_1, \lambda_3, \lambda_5)$ coincide, so $W'(\lambda_3, \lambda_5) = \pm 4q^7$.*

Proof. (i) $\mathcal{K}_\lambda \subset \mathbb{F}_{q^6}$ and $\mathfrak{q}_\lambda^{(12)} = 2\,\mathrm{Tr}_{q^6/2}$ on $\mathbb{F}_{q^6}$ (Lemma 6.3), which vanishes on $\mathcal{K}_\lambda$ (Lemma 4.2(ii)); so $S'' \neq 0$ and Proposition 6.5 applies. (ii) $\mathcal{K}_\lambda \subset \mathbb{F}_{q^3}$ is the radical of $\mathfrak{q}_\lambda^{(3)}$; if $s_3 \neq 0$ then $|s_3|^2 = 2^{3k+4}$ with $3k+4$ odd, so $\nu_3 := -s_3/4q^{3/2}$ is a primitive 8th root of unity (Lemma 2.1), all ω_j^3 equal $q^{3/2}\nu_3$ (as in Theorem 6.6), and $S'' = -4q^6\nu_3^4 = +4q^6$. (iii) Likewise with $|s_6|^2 = 2^{6k+4}$, even exponent, $\omega_j^6 = q^3\nu_6$ and $S'' = -4q^6\nu_6^2$. (iv) In

these classes $f_1 = 0$, so T'_λ is a bijection of $\mathbb{F}_q$ and Lemma 6.4 makes S'' independent of λ_1. □

Proof of Theorem 1.4. (a) is Proposition 6.5; (b) is Theorem 6.6 and Corollary 6.7; (c) is Theorem 6.9; (d) is Theorem 6.12; (e) is Theorem 6.14. □

6.9. **The phase form.** The dependence of S'' on λ_1 has a structure of its own. Fix (λ_3, λ_5) with $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{12}}$ and put

$$V_\lambda = \{c \in \mathbb{F}_{q^{12}} : T'_\lambda(c) \in \mathbb{F}_q\} \supseteq \mathbb{F}_q + \mathcal{K}_\lambda + \{\text{roots of } B_\lambda\}, \qquad \tau = \mathrm{Tr}_{q^{12}/q}\,.$$

Proposition 6.15. (i) *$T'_\lambda(V_\lambda) = \{a \in \mathbb{F}_q : \mathrm{Tr}_{q/2}(a\,\tau(x)) = 0\ \forall x \in \mathcal{K}_\lambda\}$; this is $\mathbb{F}_q$ unless $N_0 = 4$, where it is the hyperplane $\tau_0^\perp$. For $c \in V_\lambda$ one has $\tau(c) \in \mathcal{K}_\lambda \cap \mathbb{F}_q$.*

(ii) *Suppose $S''(\lambda_1^*) \neq 0$ for some λ_1^*. Then there is a well-defined function $\bar{Q} : T'_\lambda(V_\lambda) \to \mathbb{F}_2$ with*

$$S''(\lambda_1^* + a) = (-1)^{\bar{Q}(a)}\, S''(\lambda_1^*) \qquad (a \in T'_\lambda(V_\lambda)),$$

namely $\bar{Q}(a) = \frac{1}{2}\mathfrak{q}_{\lambda^*}(c)$ for any $c \in V_\lambda$ with $T'_\lambda(c) = a$, and $\bar{Q}$ is an $\mathbb{F}_2$-quadratic form with polar form $\bar{b}(a, a') = \mathrm{Tr}_{q/2}\big(\tau(c)\,a'\big)$, $T'_\lambda(c) = a$. Consequently

$$W'(\lambda_3, \lambda_5) = S''(\lambda_1^*) \sum_{a \in T'_\lambda(V_\lambda)} (-1)^{\bar{Q}(a)} \in S''(\lambda_1^*) \cdot \big(\{0\} \cup \{\pm 2^{(n+r)/2}\}\big),$$

where $n = \dim_{\mathbb{F}_2} T'_\lambda(V_\lambda) \in \{k, k-1\}$ and r is the dimension of the radical of $\bar{b}$.

(iii) *$W' = 0$ iff $\bar{Q}$ does not vanish identically on the radical of $\bar{b}$; otherwise $|W'| = 2^{(n+r)/2}|S''(\lambda_1^*)|$, and $|W'| = 2^n|S''(\lambda_1^*)|$ (i.e. $q|S''|$, or $\frac{q}{2}|S''|$ when $N_0 = 4$) iff $\bar{Q} \equiv 0$. The values of Part II ($W' \in \{0, \pm 4q^7, \pm 2q^7\}$) together with the sign counts of Table 5 mean that $\bar{b} \equiv 0$, i.e. that $\bar{Q}$ is a linear form (zero when $W' \neq 0$, nonzero when $W' = 0$), in every computed case.*

Proof. (i) T'_λ is self-adjoint for $\mathrm{Tr}_{q^{12}/2}(xy)$, so its image in $\mathbb{F}_{q^{12}}$ is $\mathcal{K}_\lambda^\perp$, and $a \in \mathbb{F}_q$ lies in it iff $\mathrm{Tr}_{q^{12}/2}(ax) = \mathrm{Tr}_{q/2}(a\tau(x)) = 0$ for all $x \in \mathcal{K}_\lambda$; Lemma 4.2(i) gives the two cases. Since T'_λ is defined over $\mathbb{F}_q$ it commutes with τ, so $T'_\lambda(\tau(c)) = \tau(T'_\lambda(c)) = 12\,T'_\lambda(c) = 0$. (ii) By Lemma 6.4, $S''(\lambda_1^* + a) = i^{-\mathfrak{q}_{\lambda^*}(c)} S''(\lambda_1^*)$; both sides are real and nonzero (Proposition 6.5 and Lemma 6.4), so $\mathfrak{q}_{\lambda^*}(c) \in \{0, 2\}$. Independence of c: another choice is $c + x$ with $x \in \mathcal{K}_\lambda$, and $\mathfrak{q}(c + x) = \mathfrak{q}(c) + \mathfrak{q}(x)$ with $\mathfrak{q}(x) = 0$ because $S''(\lambda_1^*) \neq 0$. The polar form is $\frac{1}{2} \cdot 2\,\mathrm{Tr}_{q^{12}/2}(cT'_\lambda(c')) = \mathrm{Tr}_{q/2}(\tau(c)a')$. The Gauss sum of an $\mathbb{F}_2$-quadratic form on an n-dimensional space is 0 or $\pm 2^{(n+r)/2}$. (iii) follows from (ii): the modulus 2^n requires $r = n$, and $\bar{Q}$ is then additive. □

Remark 6.16. In the classes 3^2, 6 and 5 1 ($f_1 = 0$) the form $\bar{b}$ vanishes for the trivial reason that $\tau(c) \in \mathcal{K}_\lambda \cap \mathbb{F}_q = 0$, and $\bar{Q}$ is a linear form on $\mathbb{F}_q$ which is 0 by Theorem 6.14(iv) and Theorem 6.9. In the classes with $f_1 > 0$ the vanishing of $\bar{b}$, i.e. $\mathrm{Tr}_{q/2}(\tau(c)\,T'_\lambda(c')) = 0$ for all $c, c' \in V_\lambda$, is an identity between the hexad and the polar operator that we have not been able to prove directly; it is equivalent to the statement that the sign pattern $\lambda_1 \mapsto S''(\lambda_1)/|S''(\lambda_1)|$ is a character of the additive group $T'_\lambda(V_\lambda)$, which is what Tables 5–6 show (e.g. half of the λ_1 in the class 3 2 1, where $n = k$ and $\bar{Q}$ is a nonzero linear form).

6.10. **What remains: the Brown invariants.** By Theorems 6.9–6.14 the evaluation of W' is complete in the class 5 1 and is reduced, in the other classes, to the following signs of Gauss sums of $\mathbb{Z}/4$-valued quadratic forms on small fields ("Brown invariants"):

- $N_0 \mid 4$: the sign of $s_4(\lambda) = \sum_{x \in \mathbb{F}_{q^4}} i^{\mathfrak{q}_\lambda^{(4)}(x)} \in \{0, \pm 4q^2\}$, a form on $\mathbb{F}_2^{4k}$ with four-dimensional radical;
- $N_0 \in \{3, 6\}$: the value of $s_6(\lambda)/4q^3 \in \{0, \pm 1, \pm i\}$, or, for $N_0 = 3$, whether $s_3(\lambda) \neq 0$.

Conjecture 6.17 (signs). *For all odd k and all $(\lambda_3,\lambda_5)\in\mathbb{F}_q\times\mathbb{F}_q^\times$:*

(i) *classes 1^6, 2^3: $s_4(\lambda)=+4q^2$ for all λ_1 (so $W'=4q^7$);*
(ii) *classes $2\,1^4$, $2^2 1^2$: $s_4(\lambda)=+4q^2$ for $q/2$ values of λ_1 and $-4q^2$ for the others ($W'=0$);*
(iii) *class $4\,1^2$: $s_4(\lambda)\in\{0,+4q^2\}$, nonzero for $q/2$ values ($W'=2q^7$); class $4\,2$: $s_4(\lambda)\in\{0,-4q^2\}$, nonzero for $q/2$ values ($W'=-2q^7$);*
(iv) *classes $3\,1^3$, 3^2, 6: $S''(\lambda)=+4q^6$ for all λ_1 ($W'=4q^7$); class $3\,2\,1$: $S''(\lambda)=\pm4q^6$ in equal numbers ($W'=0$).*

Equivalently, W'/q^7 is the last column of Table 1.

By Part II (quadratic-form algorithm, all pairs, $q\le 128$) the values of W' in the conjecture hold for $q\le 128$; the present L-polynomial computations confirm the finer statements about the individual $S''(\lambda_1)$ and s_m for $q=8$ (all λ) and $q=32$ (four pairs in each of the eleven classes, all λ_1; `data/witt_lpoly_k5.txt`).

Proof of Corollary 1.5. By (1),

$$q^3X=W'(0,0)+\sum_{\lambda_3\neq0}W'(\lambda_3,0)+\sum_{\lambda_5\neq0}W'(0,\lambda_5)+\sum_{\lambda_3\lambda_5\neq0}W'(\lambda_3,\lambda_5).$$

The first two terms do not involve the hexad ($\lambda_5=0$) and are known from Part II: $W'(0,0)=q^7$ (Lemma 6.15 there) and $\sum_{\lambda_3\neq0}W'(\lambda_3,0)=q^8$ (Lemma 6.16 there: $W'(\lambda_3,0)=2q^7$ for the $q/2$ values of λ_3 for which $z^3+z+\lambda_3^2$ does not have exactly one root in $\mathbb{F}_q$, and 0 otherwise). Summing the values of Conjecture 6.17 (and Theorem 6.9) against the counts of Table 3: the column $\lambda_3=0$ gives $\frac{q-2}{2}\cdot2q^7+\frac{q-2}{6}\cdot4q^7+\frac{q+1}{3}\cdot4q^7=3q^8-2q^7$, and the column $\lambda_3\neq0$ gives

$$4q^7\Big[\tfrac{(q-2)(q-8)}{720}+\tfrac{(q-2)(q-8)}{48}+\tfrac{2(q+1)(q-2)}{18}+\tfrac{(q+1)(q-2)}{6}\Big]$$
$$+2q^7\tfrac{(q-2)(q-4)}{8}-2q^7\tfrac{q(q-2)}{8}-q^7\tfrac{q^2+1}{5}=q^9-3q^8+q^7;$$

the total is $q^7+q^8+(3q^8-2q^7)+(q^9-3q^8+q^7)=q^9+q^8=q^3(q^6+q^5)$. (At $q=8$ and $q=32$ the four parts are $1,8,22,41$ and $1,32,94,929$ in units of q^7, with totals $72=q^2+q$ and $1056=q^2+q$.) The contribution $-q^7(q^2+1)/5$ of the class $5\,1$ uses only Theorem 6.9 and the count $(q^2+1)/5$. □

6.11. **Even k.** The parity of k entered the proofs above only through Lemma 2.1, i.e. through the parity of the exponent $mk+f_m$ of $|s_m|^2=2^{mk+f_m}$. Everything else holds for all k, and the arguments give the opposite signs for even k.

Theorem 6.18. *Let k be even. Then Theorem 1.1, Theorem 1.2, Theorem 1.3, Lemma 4.2, Proposition 6.5, Theorem 6.6, Theorem 6.12, Theorem 6.14*(i),(iii),(iv) *and Proposition 6.15 hold verbatim, and:*

(i) *in the class $5\,1$: $s_1(\lambda)=\eta\sqrt{q}$ with $\eta\in\{\pm1,\pm i\}$, the L-polynomial is $(1-\eta^5q^{5/2}T^5)/(1-\eta q^{1/2}T)$, $S''(\lambda)=+q^6$ for every λ_1 and $W'(\lambda_3,\lambda_5)=+q^7$;*
(ii) *in the classes with $N_0=3$: if $s_3(\lambda)\neq0$ then $S''(\lambda)=-4q^6$.*

Proof. Same proofs; in Theorem 6.9, $|s_1|^2=2^k$ with k even gives $\eta\in\{\pm1,\pm i\}$, $\eta^4=1$ and $S''=q^6\eta^{12}=q^6$; in Theorem 6.14(ii), $|s_3|^2=2^{3k+4}$ has even exponent, so $\nu_3\in\{\pm1,\pm i\}$, $\nu_3^4=1$ and $S''=-4q^6$. □

This is the mechanism behind the sign $(-1)^{k+1}$ in the conjectural formula $X=(-1)^{k+1}q^6+q^5$ of Part II. The data for even k (Table 4; `sextic_types.py 2 4 6 8`, `witt_lpoly.py 2, 4`) show two further phenomena. First, the class counts are again

polynomials in q with the Chebotarev leading terms, but they depend on k modulo 4, i.e. on whether $\mathbb{F}_q$ contains the fifth roots of unity ($q \equiv 1 \bmod 5$ iff $4 \mid k$): only the classes 1^6, $2^2 1^2$ and $5\,1$ (and the pure column) are affected. Second, at $q = 4$ and $q = 16$ (all pairs, $\sum_{\lambda_1} S'' = W'$ in every case) the values of W' in the ten classes that occur are the negatives of the odd-k values: $5\,1$: $+q^7$; $3\,1^3$, 3^2, 2^3, 6: $-4q^7$; $4\,1^2$: $-2q^7$; $4\,2$: $+2q^7$; $2\,1^4$, $2^2 1^2$, $3\,2\,1$: 0; the classes $5\,1$, $3\,1^3$, 3^2 as Theorem 6.18 predicts. The moment identities of Theorem 1.3 hold for even k as well (their proofs do not use the parity), and the polynomials of Table 4 satisfy them; the total of the mixed column is $(q-1)^2$ and that of the pure column is $q-1$ in both residue classes. If the class 1^6 (which first occurs at $q = 64$) also has $W' = -4q^7$, the tables give $\sum_{\lambda_3\lambda_5\neq 0} W' = -q^9 + 3q^8 - 3q^7$ and $\sum_{\lambda_5\neq 0} W'(0,\lambda_5) = -(q-2)q^7$ in both residue classes; with $W'(0,0) = -q^7$ and $\sum_{\lambda_3\neq 0} W'(\lambda_3,0) = -(q-2)q^7$ (the pure cubic sums at $q = 4, 16$; cf. Part II, Lemma 6.16) the total is $-q^9 + q^8$, i.e. $X = -q^6 + q^5$, the even-k formula of Part II. At $q = 4$ and $q = 16$ the four parts are exactly $-1, -2, -2, -7$ and $-1, -14, -14, -211$ in units of q^7.

TABLE 4. Even k: the number of (λ_3, λ_5) with B_λ of each class and the value of W'/q^7. The polynomials are fitted to the exact counts at $q = 4, 64$ ($k \equiv 2 \bmod 4$) and $q = 16, 256$ ($k \equiv 0 \bmod 4$) with the leading coefficient $|C|/720$ imposed, so each is determined by two data points and is conjectural; they agree with the counts at $q = 1024$ for the classes present there (see the text). The last column gives W'/q^7 at $q = 4, 16$ for the classes that occur there (`witt_lpoly.py 2, 4`, all pairs); the class 1^6 first occurs at $q = 64$.

class	$\lambda_3 \neq 0$, $k \equiv 2$ (4)	$\lambda_3 \neq 0$, $k \equiv 0$ (4)	$\lambda_3 = 0$, $k \equiv 2$	$\lambda_3 = 0$, $k \equiv 0$	W'/q^7
1^6	$(q-4)(q-28)/720$	$(q-16)^2/720$	$(q-4)/60$	$(q-16)/60$	–
$2\,1^4$	$q(q-4)/48$	$q(q-4)/48$	0	0	0
$2^2\,1^2$	$(q-4)^2/16$	$q(q-8)/16$	$(q-4)/4$	$q/4$	0
2^3	$q(q-4)/48$	$q(q-4)/48$	0	0	-4
$3\,1^3$	$(q-1)(q-4)/18$	$(q-1)(q-4)/18$	0	0	-4
3^2	$(q-1)(q-4)/18$	$(q-1)(q-4)/18$	$(q-1)/3$	$(q-1)/3$	-4
$3\,2\,1$	$q(q-1)/6$	$q(q-1)/6$	0	0	0
$4\,1^2$	$q^2/8$	$q^2/8$	0	0	-2
$4\,2$	$q^2/8$	$q^2/8$	0	0	2
$5\,1$	$(q-3)(q+1)/5$	$(q-1)^2/5$	$2(q+1)/5$	$2(q-1)/5$	1
6	$q(q-1)/6$	$q(q-1)/6$	0	0	-4

7. Computations

All computations are exact. The scripts are in the directory `code/` of the repository; $\mathbb{F}_{q^{12}}$ is built as in Part II.

`hexad_search.py` computes $\mathcal{K}_\lambda$ as an $\mathbb{F}_2$-nullspace, enumerates the 168 frames and 28 translation classes of hexads, and finds the Frobenius-stable classes and their normalised sextics. For $q = 8, 32$: every pair with $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{12}}$ has at least one stable class; the normalised sextic of exactly one stable class is B_λ/λ_5^2 for every pair (`hexad_fit.py` found the coefficients by solving linear systems over $\mathbb{F}_q$ for the exponents of λ_3, λ_5); its factorisation type is the class of Table 1. The numbers of stable classes and the types of σ with respect to them are exactly those predicted by Proposition 4.3 (Table 2, last column; `stable_stats.py`): e.g. every pair of class $2^2 1^2$ or 2^3 has eight stable classes, six of type $2^2 1^2$ and two of type 2^3, and every pair of order 4 has two, one of each type. Thus the orbit structure alone does not determine the class, while the canonical hexad does.

`sextic_identity.py` verifies Lemma 3.1, the membership of $L'_\lambda(r+s)$ in the ideal $(B_\lambda(r), B_\lambda(s))$ by a Gröbner basis over $\mathbb{F}_2[r, s, \lambda_3, \lambda_5]$, and the kernel description for all pairs at $q = 8, 32$.

`sextic_types.py` produced Table 3 ($q \le 512$).

`hexad_signs.py` computes, for every pair at $q = 8, 32$, the roots as the solutions of $B_\lambda = 0$ in the coset $T'^{-1}_\lambda(1)$ (confirming Theorem 1.1(d)), their traces, the values of the $\mathbb{F}_2$-form Q' of Part II on them, and the radical of Q'. `hexad_traces.py` checks Lemma 4.2 and the existence of a root with $\mathrm{Tr}_{q^{12}/2}(r) = 0$ in every class ($q = 8, 32$: all true).

`witt_lpoly.py` computes the Witt sums s_m, $m \le 4$ (vectorised Witt traces over $\mathbb{F}_{2^{mk}}$), the eigenvalues by Newton's identities, the predicted $S'' = -\sum \omega_j^{12}$, and compares $\sum_{\lambda_1} S''$ with the exact W' of Part II; it produced Tables 5 and 6 and the data of Figure 4. At $q = 8$ (all 56 pairs, 448 values of λ; 2.5 min) the sums of the predictions agree with W' for every pair, the ζ_j are 24th roots of unity when $\mathcal{K}_\lambda \subset \mathbb{F}_{q^{12}}$ and 40th roots in the class $5\,1$, every S'' is in $\{0, \pm 4q^6\}$ or equals $-q^6$, and the sign pattern is that of Conjecture 6.17; the same holds at $q = 32$ for four pairs in each of the eleven classes (1408 values of λ, about two hours), and at $q = 4$ and $q = 16$ (all pairs) with the even-k signs of §6.11.

TABLE 5. Normalised Frobenius eigenvalues $\zeta_j = \omega_j/\sqrt{q} = e^{2\pi i a_j/N}$ of the rank-four Witt sheaf at $q = 8$, by class and value of $S''(\lambda_1)/4q^6$ ($N = 24$, except $N = 40$ in the class $5\,1$, where $S''/q^6 = -1$); every multiset $\{a_j\}$ that occurs is listed, with the number of $(\lambda_3, \lambda_5, \lambda_1)$ in brackets. The multisets with $S'' = +4q^6$ consist of odd exponents ($\zeta^{12} = -1$), those with $-4q^6$ of even exponents ($\zeta^{12} = +1$), those with 0 of two of each (Corollary 6.7).

class	$S''/4q^6$	multisets $\{a_j\}$ (mod N)
$3\,1^3$	$+1$	$\{1,5,13,17\}$ [9], $\{7,11,19,23\}$ [9], $\{5,5,13,13\}$ [3], $\{11,11,19,19\}$ [3]
2^3	$+1$	$\{3,9,15,21\}$ [6], $\{3,15,15,15\}$ [1], $\{9,9,9,21\}$ [1]
3^2	$+1$	$\{1,9,9,17\}$ [9], $\{7,15,15,23\}$ [9], $\{3,3,11,19\}$ [3], $\{5,13,21,21\}$ [3]
6	$+1$	$\{3,11,15,19\}$ [36], $\{5,9,13,21\}$ [36], $\{1,9,17,21\}$ [12], $\{3,7,15,23\}$ [12]
$2^2\,1^2$	-1	$\{0,6,12,18\}$ [12]
	$+1$	$\{3,3,15,15\}$ [3], $\{3,9,9,15\}$ [3], $\{9,9,21,21\}$ [3], $\{9,15,15,21\}$ [3]
$3\,2\,1$	-1	$\{2,10,14,22\}$ [18], $\{4,8,16,20\}$ [18]
	$+1$	$\{5,11,13,19\}$ [18], $\{5,7,13,23\}$ [9], $\{1,11,17,19\}$ [9]
$4\,1^2$	$+1$	$\{3,9,15,21\}$ [24]
	0	$\{6,9,15,18\}$ [12], $\{0,3,9,12\}$ [6], $\{0,12,15,21\}$ [6]
$4\,2$	-1	$\{0,6,12,18\}$ [24]
	0	$\{0,9,12,15\}$ [12], $\{3,6,9,18\}$ [6], $\{6,15,18,21\}$ [6]
$5\,1$	$-\frac{1}{4}$	$\{3,11,19,27\}$ [39], $\{13,21,29,37\}$ [39], $\{1,9,17,33\}$ [13], $\{7,23,31,39\}$ [13]

TABLE 6. The small-field Gauss sums at $q = 8$: $s_m/q^{m/2}$ for $m = 1, \dots, 4$, written as "f_m:b" for $2^{f_m/2}\zeta_8^b$, or 0, by class and value of $S''/4q^6$, with multiplicities in brackets. One reads off Theorem 6.9(iii) ($s_m = s_1^m$ with b_1 odd in the class $5\,1$), Theorem 6.12 ($S''/4q^6 = s_4/4q^2$ when $N_0 \mid 4$: $b_4 = 0$ for $+1$, $b_4 = 4$ for -1, $s_4 = 0$ for 0) and Theorem 6.14(ii) (b_3 odd whenever $s_3 \neq 0$ with full radical).

class	$S''/4q^6$	(s_1, s_2, s_3, s_4) normalised
$5\,1$	$-\frac{1}{4}$	(0:b, 0:2b, 0:3b, 0:4b): $b = 1$ [39], $b = 3$ [13], $b = 5$ [13], $b = 7$ [39]
$4\,1^2$	$+1$	$(0, 0, 0, 4{:}0)$ [24]
	0	$(1{:}0, 2{:}0, 1{:}4, 0)$ [12], $(1{:}2, 2{:}4, 1{:}2, 0)$ [6], $(1{:}6, 2{:}4, 1{:}6, 0)$ [6]
$4\,2$	-1	$(0, 0, 0, 4{:}4)$ [24]
	0	$(1{:}0, 2{:}4, 1{:}4, 0)$ [12], $(1{:}2, 2{:}0, 1{:}2, 0)$ [6], $(1{:}6, 2{:}0, 1{:}6, 0)$ [6]
2^3	$+1$	$(0, 0, 0, 4{:}0)$ [6], $(2{:}1, 4{:}6, 2{:}3, 4{:}0)$ [1], $(2{:}7, 4{:}2, 2{:}5, 4{:}0)$ [1]
$2^2\,1^2$	-1	$(0, 0, 0, 4{:}4)$ [12]
	$+1$	$(0, 4{:}2, 0, 4{:}0)$ [3], $(0, 4{:}6, 0, 4{:}0)$ [3], $(2{:}1, 0, 2{:}3, 4{:}0)$ [3], $(2{:}7, 0, 2{:}5, 4{:}0)$ [3]
$3\,1^3$	$+1$	$(0, 2{:}2, 0, 2{:}4)$ [9], $(0, 2{:}6, 0, 2{:}4)$ [9], $(2{:}1, 2{:}2, 4{:}7, 2{:}4)$ [3], $(2{:}7, 2{:}6, 4{:}1, 2{:}4)$ [3]
3^2	$+1$	$(0{:}1, 0{:}6, 4{:}3, 0{:}0)$ [9], $(0{:}7, 0{:}2, 4{:}5, 0{:}0)$ [9], $(0{:}3, 0{:}2, 4{:}1, 0{:}0)$ [3], $(0{:}5, 0{:}6, 4{:}7, 0{:}0)$ [3]
6	$+1$	$(0{:}1, 0{:}6, 2{:}7, 0{:}0)$ [36], $(0{:}7, 0{:}2, 2{:}1, 0{:}0)$ [36], $(0{:}3, 0{:}2, 2{:}5, 0{:}0)$ [12], $(0{:}5, 0{:}6, 2{:}3, 0{:}0)$ [12]
$3\,2\,1$	-1	$(0, 2{:}0, 0, 2{:}0)$ [18], $(0, 2{:}4, 0, 2{:}0)$ [18]
	$+1$	$(1{:}0, 0, 3{:}0, 2{:}4)$ [18], $(1{:}2, 0, 3{:}6, 2{:}4)$ [9], $(1{:}6, 0, 3{:}2, 2{:}4)$ [9]

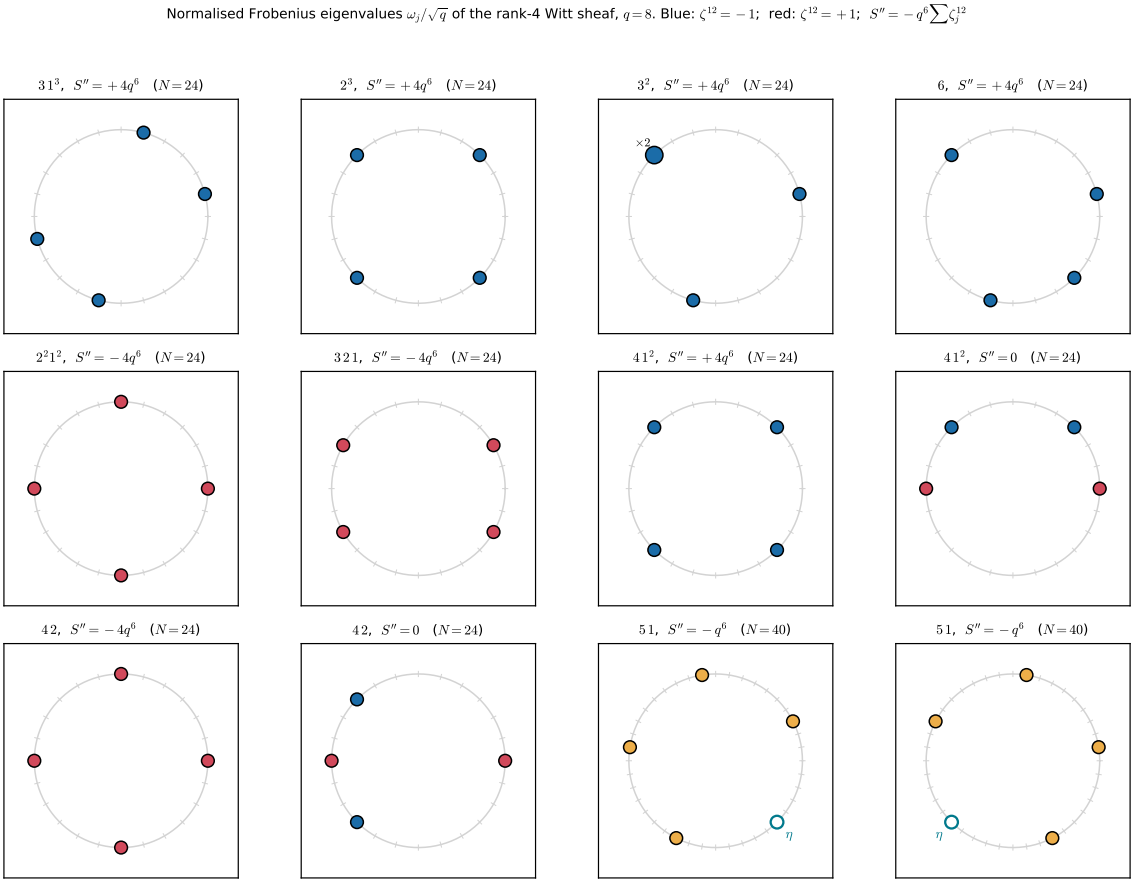


FIGURE 4. Normalised Frobenius eigenvalues of the Witt sheaf for one representative of each (class, value of S'') at $q = 8$ (Table 5). Blue points have $\zeta^{12} = -1$, red points $\zeta^{12} = +1$; $S'' = -q^6 \sum \zeta_j^{12}$. In the class $5\,1$ the four eigenvalues are the nontrivial 5th-root translates of the primitive 8th root $\eta = s_1/\sqrt{q}$ (open circle), as Theorem 6.9 predicts.

Remark 7.1 (reading the tables)**.** Several of the regularities in Tables 5–6 are theorems proved above, and the others are precise targets. (1) In the class 2^3, and in the class $2^2 1^2$ when $S'' = +4q^6$, s_2 is 0 or $\pm 4qi$, never $\pm 4q$: indeed if $s_2 = \pm 4q$ then all $\zeta_j^2 = \mp 1$, $\zeta_j^4 = 1$ and $S'' = -4q^6$. (2) In the classes of order 4 with $S'' = 0$ one has $s_1 \neq 0$ (radical $\langle \tau_0 \rangle$, $f_1 = 1$), while for $S'' \neq 0$, $s_1 = s_2 = s_3 = 0$: the hyperplane of Theorem 6.12 is the complement of the hyperplane on which $\mathfrak{q}_\lambda^{(1)}$ vanishes on τ_0. We have not proved this complementarity. (3) In the class $3\,1^3$, $s_4 = -2q^2$ always ($f_4 = 2$) and s_3 vanishes for three quarters of the λ_1; when it does not, b_3 is odd, as Theorem 6.14(ii) requires. (4) In the classes 3^2 and 6 the first Gauss sum s_1 has trivial radical and is therefore $\sqrt{q}$ times a primitive 8th root of unity, exactly as in the class $5\,1$; the difference is that $f_3 > 0$ there, so the rigidity argument of Theorem 6.9 does not apply.

8. Status and open problems

Table 7. What is proved, what is verified, what is conjectural.

statement	status	where
$\mathcal{K}_\lambda = \{0\} \cup \{r_i + r_j\}$ for the roots of $B_\lambda = uA_\lambda + 1$; $T'_\lambda(r_i) = 1$; deformation $uA_\lambda + \delta$	proved, all k	Thms 1.1, 3.7
Frobenius class = factorisation type of B_λ; orbit structures, N_0, f_m	proved, all k	Thm 1.2
Number of σ-stable symplectic forms of each type $= 28\|t\|/\|C\|$	proved; data $q = 8, 32$	Prop. 4.3
Moments $(q-1)^2$, $(q-2)^2$, $q(q-2)$; pair lemma $\mathrm{Tr}(r+s) = 0$	proved, all k	Thm 1.3
Class counts, k odd (Table 3)	verified $q \leq 512$; $\lambda_3 = 0$ proved	Conj. 5.3
Class counts, k even (Table 4)	fitted to $q \leq 256$	§6.11
$S''(\lambda) \in \mathbb{Z}$, $\in \{0, \pm 4q^6\}$ or $\pm q^6$; $S''(\lambda_1) = S''(\lambda_1 + 1)$	proved, all k	Prop. 6.5
$\|\omega_j\| = \sqrt{q}$, $\omega_j^{4N_0}$ all equal, ζ_j roots of unity	proved, all k	Thm 6.6
Class $5\,1$: L-polynomial, $S'' = (-1)^{k+1}q^6$, $W' = (-1)^{k+1}q^7$	proved, all k	Thms 6.9, 6.18
$N_0 \mid 4$: $S'' = q^4 s_4$; nonvanishing pattern (q or $q/2$ values)	proved, all k	Thm 6.12
$N_0 \in \{3, 6\}$: $S'' = \pm 4q^6$ for all λ_1; $+$ if $s_3 \neq 0$ (k odd)	proved	Thm 6.14
Sign pattern in λ_1 is an $\mathbb{F}_2$-quadratic form on $T'_\lambda(V_\lambda)$	proved	Prop. 6.15
Signs of s_4 ($N_0 \mid 4$), s_6 ($N_0 \in \{3, 6\}$); W' by class	W' verified $q \leq 128$; s_m, $S''(\lambda_1)$ at $q = 8, 32$	Conj. 6.17
$X = q^6 + q^5$ (k odd)	$\Leftarrow$ Conj. 5.3 + Conj. 6.17	Cor. 1.5

Open Problem 1. Prove Table 3: the factorisation statistics of the family $u^6 + au^2 + bu + b^2$ over $\mathbb{F}_q$, $q = 2^k$, k odd. The four relations of Corollary 5.1 are proved; the remaining moments are point counts of explicit curves which the table predicts to be polynomial.

Open Problem 2. Evaluate the Brown invariants of §6.10: the sign of $s_4(\lambda)$ for $N_0 \mid 4$ and the value of $s_6(\lambda)/4q^3$ for $N_0 \in \{3, 6\}$ (Conjecture 6.17). The forms are explicit: $\mathfrak{q}_\lambda^{(4)}$ on $\mathbb{F}_{q^4}$ with radical $\mathcal{K}_\lambda$, and $\mathfrak{q}_\lambda^{(6)}$ on $\mathbb{F}_{q^6}$; by the shift calculus the value for one λ_1 in each coset of $T'_\lambda(\mathbb{F}_q)$ suffices. Together with Open Problem **Open Problem 1.** this would complete the proof of $X = q^6 + q^5$ for odd k.

Open Problem 3. Extend to even k. The hexad theorem and Theorems 6.6, 6.12, 6.14(i),(iii),(iv) hold verbatim; in Theorem 6.9 the parity enters through Lemma 2.1 and

gives $S'' = +q^6$, in agreement with the sign $(-1)^{k+1}$ of Part II; the class counts must change since the Gold quintic has different statistics for even k.

Open Problem 4. Interpret the hexad. The six roots of B_λ are attached to the supersingular curve $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3$ (they do not depend on λ_1), and $\mathrm{Sp}_4(\mathbb{F}_2) \cong S_6$ is the automorphism group of the 2-torsion of a principally polarised abelian surface with its Weil pairing, the six roots playing the role of the Weierstrass points of a genus-two curve. In characteristic 2 the Jacobian of a supersingular genus-two curve has no étale 2-torsion, and $(\mathcal{K}_\lambda, \omega)$ should be a substitute: the 2-torsion of a lift, or an étale piece of a Dieudonné-type object attached to the rank-four Witt sheaf. A conceptual explanation along these lines would presumably also explain the exact class counts and the Brown invariants.

Open Problem 5. Prove that the phase form $\bar{Q}$ of Proposition 6.15 is linear, i.e. $\mathrm{Tr}_{q/2}\big(\tau(c)\, T'_\lambda(c')\big) = 0$ for all $c, c' \in V_\lambda$; and prove the complementarity observed in the classes of order 4 (Remark after Table 6): $S''(\lambda_1) \neq 0$ iff $s_1(\lambda_1) = 0$.

Open Problem 6. Explain the dependence of the even-k class counts on k modulo 4 (Table 4) and prove them; the odd-k counts do not depend on k modulo 4 although $q \equiv 2, 3 \pmod 5$ alternates.

Open Problem 7. The deformation $uA_\lambda(u) + \delta$ (Theorem 3.7) connects the quadratic sums W of A_6 ($\delta = 0$) and the order-4 sums W' of X ($\delta = 1$). For intermediate δ the sums have no character-sum meaning; but the family of sextics does, and its monodromy ($S_5 \subset S_6$ at the boundary) may explain why the quintic statistics of Part II (three classes) specialise the sextic statistics (eleven classes) exactly as in the column $\lambda_3 = 0$ of Table 3.

Acknowledgements

The computations were carried out with NumPy and SymPy; the search that found B_λ (§7) was designed and run in an interactive session with Claude (Anthropic), which also assisted with the drafting of the manuscript.

Appendix A. All pairs at $q = 8$

Table 8 lists, for $q = 8$, every $(\lambda_3, \lambda_5) \in \mathbb{F}_8 \times \mathbb{F}_8^\times$ with the factorisation type of B_λ over $\mathbb{F}_8$ and the exact value of $W'(\lambda_3, \lambda_5)/q^7$ from Part II. Here $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$. Among the 49 pairs with $\lambda_3 \neq 0$ the classes $3\,1^3$, $2^2 1^2$, 3^2, $4\,1^2$, $4\,2$, $3\,2\,1$, $5\,1$, 6 occur $3, 3, 3, 3, 6, 9, 13, 9$ times, and among the 7 pairs with $\lambda_3 = 0$ the classes $4\,1^2$, 2^3, 6 occur $3, 1, 3$ times, in agreement with Table 3 at $q = 8$ (where $(q-2)(q-8) = 0$ kills the classes 1^6, $2\,1^4$ and the mixed part of 2^3). The values of W' are constant on the classes, as Theorem 1.4 and Conjecture 6.17 assert; the total $\sum W' = q^9 + q^8 - W'(0,0) = q^9 + q^8 - q^7$ over the 56 pairs gives $X = q^6 + q^5 = 294912$.

TABLE 8. $q = 8$: all pairs (λ_3, λ_5), the class of B_λ and W'/q^7 (generated by `appendix_table_k3.py`).

λ_3	λ_5	class	W'/q^7	λ_3	λ_5	class	W'/q^7
α	$\alpha^2+\alpha+1$	$2^2 1^2$	0	α	$\alpha+1$	$4\,2$	-2
α^2	$\alpha+1$	$2^2 1^2$	0	α^2	α^2+1	$4\,2$	-2
$\alpha^2+\alpha$	α^2+1	$2^2 1^2$	0	$\alpha^2+\alpha$	$\alpha^2+\alpha+1$	$4\,2$	-2
0	1	2^3	4	1	1	$5\,1$	-1
$\alpha+1$	$\alpha^2+\alpha$	$3\,1^3$	4	α	α	$5\,1$	-1
α^2+1	α	$3\,1^3$	4	$\alpha+1$	1	$5\,1$	-1
$\alpha^2+\alpha+1$	α^2	$3\,1^3$	4	$\alpha+1$	$\alpha+1$	$5\,1$	-1
α	1	$3\,2\,1$	0	$\alpha+1$	$\alpha^2+\alpha+1$	$5\,1$	-1
α	α^2+1	$3\,2\,1$	0	α^2	α^2	$5\,1$	-1
$\alpha+1$	α^2	$3\,2\,1$	0	α^2+1	1	$5\,1$	-1
α^2	1	$3\,2\,1$	0	α^2+1	$\alpha+1$	$5\,1$	-1
α^2	$\alpha^2+\alpha+1$	$3\,2\,1$	0	α^2+1	α^2+1	$5\,1$	-1
α^2+1	$\alpha^2+\alpha$	$3\,2\,1$	0	$\alpha^2+\alpha$	$\alpha^2+\alpha$	$5\,1$	-1
$\alpha^2+\alpha$	1	$3\,2\,1$	0	$\alpha^2+\alpha+1$	1	$5\,1$	-1
$\alpha^2+\alpha$	$\alpha+1$	$3\,2\,1$	0	$\alpha^2+\alpha+1$	α^2+1	$5\,1$	-1
$\alpha^2+\alpha+1$	α	$3\,2\,1$	0	$\alpha^2+\alpha+1$	$\alpha^2+\alpha+1$	$5\,1$	-1
α	α^2	3^2	4	0	α	6	4
α^2	$\alpha^2+\alpha$	3^2	4	0	α^2	6	4
$\alpha^2+\alpha$	α	3^2	4	0	$\alpha^2+\alpha$	6	4
0	$\alpha+1$	$4\,1^2$	2	1	$\alpha+1$	6	4
0	α^2+1	$4\,1^2$	2	1	α^2+1	6	4
0	$\alpha^2+\alpha+1$	$4\,1^2$	2	1	$\alpha^2+\alpha+1$	6	4
$\alpha+1$	α	$4\,1^2$	2	α	$\alpha^2+\alpha$	6	4
α^2+1	α^2	$4\,1^2$	2	$\alpha+1$	α^2+1	6	4
$\alpha^2+\alpha+1$	$\alpha^2+\alpha$	$4\,1^2$	2	α^2	α	6	4
1	α	$4\,2$	-2	α^2+1	$\alpha^2+\alpha+1$	6	4
1	α^2	$4\,2$	-2	$\alpha^2+\alpha$	α^2	6	4
1	$\alpha^2+\alpha$	$4\,2$	-2	$\alpha^2+\alpha+1$	$\alpha+1$	6	4

APPENDIX B. ALGORITHMS AND RUN TIMES

Fields. $\mathbb{F}_{q^{12}}$ is represented as $\mathbb{F}_q[z]/(m(z))$ with m irreducible of degree 12 over $\mathbb{F}_q$ (fqlib tables of Part II), and every $\mathbb{F}_2$-linear map ($x \mapsto x^2$, $x \mapsto x^{1/2}$, multiplication by a constant, $\mathrm{Tr}_{q^{12}/q}$, T'_λ) is stored as a $12k \times 12k$ matrix over $\mathbb{F}_2$; $\mathcal{K}_\lambda$ is the nullspace of the matrix of $L'_\lambda = (T'_\lambda)^4$, computed by Gaussian elimination over $\mathbb{F}_2$.

Hexad search. The 168 frames of $\mathcal{K}_\lambda$ are the 4-subsets of $\mathcal{K}_\lambda \setminus 0$ of rank 4 completed by their sum (there are $\binom{15}{4} = 1365$ subsets, 840 of rank 4, and each frame arises from its 5 four-subsets); a frame F gives the hexad $H = \{0\} \cup F$, whose translation class is tested for σ-stability by checking whether $\sigma(H) + h \in \{H + v : v \in \mathcal{K}_\lambda\}$ for some h. For a stable class the polynomial $\prod_{h \in H}(u + h)$ is computed, the translate killing the u^4-coefficient is taken, and the coefficients are tested for membership in $\mathbb{F}_q$. Cost per pair: $O(168 \cdot 16)$ polynomial evaluations in $\mathbb{F}_{q^{12}}$; run times a few seconds ($q = 8$), 5 min ($q = 32$), several hours ($q = 128$, 16256 pairs).

Class counts. `sextic_types.py` computes $\gcd(B_\lambda, u^{q^m} - u)$, $m = 1, \ldots, 6$, by repeated squaring modulo B_λ over $\mathbb{F}_q$; run times under a second for $q = 8, 32$, 11 s for $q = 128$ and 5 min for $q = 512$.

Witt sums. For $m \le 4$ the field $\mathbb{F}_{2^{mk}}$ is given by log/exp tables; the Witt trace $\mathsf{T}_{mk}(x)$ of all $2^{mk} - 1$ nonzero x at once is accumulated by the vectorised recursion $(s_0, s_1) \leftarrow (s_0 + c,\ s_1 + s_0 c)$ over the mk conjugates $c = x, x^2, \ldots$ (this is (6) evaluated by Horner-type accumulation of the Witt sum), and $2\,\mathrm{Tr}_{q^m/2}\, f_\lambda(x)$ is added; the Gauss

sum is then a histogram of the four values. Cost $O(mk\,2^{mk})$ per (λ, m); for $q = 8$ all 448 values of λ take 2.5 min and for $q = 16$ all 3840 values about 50 min; at $q = 32$ the sums with $m = 4$ run over $\mathbb{F}_{2^{20}}$ and the sample of four pairs per class (1408 values of λ) took about two hours. Newton's identities over $\mathbb{Q}(i)$ give the e_j of the ω_j from $s_1, \dots, s_4$; the roots are computed numerically and recognised as $\sqrt{q}$ times roots of unity (exponents on a common denominator 120), and $S'' = -\sum \omega_j^{12}$ is evaluated exactly from the power sums.

Reference values. The exact $W'(\lambda_3, \lambda_5)$ are those of Part II, obtained from the $\mathbb{F}_2$-quadratic form Q' on $\mathbb{F}_{q^{12}}$ by the radical algorithm (`legendre_d6_qf.py`); the agreement $\sum_{\lambda_1} S''(\lambda) = W'$ in every case is therefore a check of two independent computations (Witt sums over $\mathbb{F}_{q^m}$, $m \le 4$, against an $\mathbb{F}_2$-form over $\mathbb{F}_{q^{12}}$) as well as of the L-polynomial principle.

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